

# Diagnostic Overreaction in Inflation Expectations: Evidence from China's Pork Supply Crisis

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April 2026

## Abstract

Households systematically overpredict inflation after salient food-price shocks. I exploit the 2018–2019 African Swine Fever outbreak in China, which reduced the national hog herd by 30–55 percent and generated sharp cross-province variation in meat prices. Combining province-level meat CPI data with four waves of the China Family Panel Studies (2016–2022), I show that a one-standard-deviation meat-price shock raises household forecast errors by 1.84 percentage points, roughly three-quarters of average headline inflation. A signal-extraction model calibrated to the CPI weight of meat bounds the rational forecast error at less than one-twentieth of the estimated response. The excess is consistent with diagnostic overreaction: households treat a transitory, mean-reverting shock as informative about persistent inflation. Placebo tests confirm that grain shocks of comparable CPI weight produce no such pattern. The findings imply that salient relative-price movements can durably distort the inflation expectations that anchor consumption and wage-setting decisions.

**JEL Codes:** D84, E31, E71

**Keywords:** inflation expectations, overreaction, diagnostic expectations, salience, household surveys, China

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# 1 Introduction

Households exposed to large pork-price shocks systematically overpredict future inflation. Using province-level meat CPI variation from the National Bureau of Statistics matched to four waves of the China Family Panel Studies (2016–2022), I estimate that a one-unit increase in the cumulative meat log-price shock generates a forecast error of  $-7.61$  percentage points under the preferred specification with province and region-by-wave fixed effects (wild cluster bootstrap  $p = 0.005$ , 31 province clusters). The negative sign means that households in high-shock provinces expected more inflation than materialised. A signal-extraction model calibrated to the actual persistence of meat-price shocks bounds the rational forecast error at  $|\beta_3^R| \leq \omega \approx 0.05$  per unit of the shock; even after accounting for time aggregation between the model and the empirical object, the most generous rational bound is 1.20. The estimated coefficient exceeds this bound by a factor of six, and no combination of persistence or CPI-weight assumptions within the plausible range closes the gap. The excess is consistent with diagnostic overreaction ([Bordalo, Gennaioli, and Shleifer](#); [Bordalo et al.](#)).

Why does this matter for macroeconomics? Inflation expectations anchor wage bargaining, consumption timing, and portfolio allocation. If a large share of the population systematically overpredicts inflation in response to salient relative-price movements, the expectation wedge feeds back into real activity through precautionary saving and consumption compression. In China, where pork accounts for roughly 60 percent of household meat consumption and food carries an official CPI weight near 28 percent, the scope for salience-driven belief distortion is larger than in economies with lower food shares or more diversified protein markets. The 2018–2019 African Swine Fever (ASF) episode, which reduced the national hog herd by an estimated 30–55 percent, provides sharp cross-province variation in the size of the salience signal.

The identification rests on a three-equation system that separates the belief channel from the price-level channel. The primary test is a reduced-form regression of household forecast errors on the province-level meat-price shock (Equation 3). Two decomposition equations ask whether the shock raises household expectations (Equation 1) and whether it predicts forward headline CPI (Equation 2). The expectation response is positive and statistically significant at the five-percent level under the preferred province-plus-

region-by-wave specification ( $\hat{\beta}_1 = 0.119$ ,  $p = 0.019$ ). The forecast-error result is the more powerful test because it measures the joint outcome of expectations minus realised inflation directly, but the significant expectation channel confirms that the excess forecast error comes from beliefs, not from an unexpected decline in realised CPI. The treatment is measured by the NBS and merged from outside the CFPS survey, so the household's current expectation never appears on both sides of any equation.

I make three contributions. The forecast-error test avoids the mechanical negative correlation in standard revision-error designs by using an external, supply-driven price signal rather than within-survey belief revisions. A calibrated signal-extraction benchmark then separates behavioural overreaction from rational forecast errors due to transitory-shock uncertainty: the estimated forecast error exceeds the rational bound under any persistence assumption. The commodity placebo test disciplines the mechanism. Meat shocks generate forecast errors while grain shocks of comparable CPI weight do not, consistent with salience-driven rather than rational updating.

The paper connects to three literatures. Diagnostic expectations ([Bordalo, Gennaioli, and Shleifer](#); [Bordalo et al.](#)) predict that agents overweight signals made representative by recent experience; the forecast-error pattern I document is a direct counterpart of this prediction. Rational inattention ([Sims](#); [Maćkowiak and Wiederholt](#)) predicts selective attention to informative signals; the commodity asymmetry is consistent with disproportionate attention to high-frequency prices. Experience-based learning ([Malmendier and Nagel](#); [Kuchler and Zafar](#)) predict that personal purchase experience shapes expectations more than official statistics; the meat channel operates through precisely these signals.

Three limits apply. The CFPS is biennial, so I cannot trace within-year adjustment dynamics. Province-level meat CPI conflates supply and demand movements; ASF provides a plausible supply instrument, but the treatment is an imperfect proxy for the supply component alone. With 31 province clusters, inference throughout relies on the wild cluster bootstrap.

The remainder of the paper proceeds as follows. Section 2 provides institutional background. Section 3 develops the conceptual framework. Section 4 describes the data. Section 5 presents the empirical strategy. Section 6 reports main results. Section 7 presents commodity placebo evidence. Section 8 contains robustness checks. Section 9

discusses alternative interpretations and policy implications. Section 10 concludes.

## 2 Institutional Background and Related Literature

### 2.1 Food Prices in Chinese Inflation

Food is the single largest category in the Chinese CPI basket. The National Bureau of Statistics weights food and non-alcoholic beverages at approximately 28 percent of the overall index, roughly twice the comparable share in high-income economies. This weight reflects both the expenditure share of food in Chinese household budgets and the NBS’s plutocratic aggregation method. Because food expenditure shares decline with income (Engel’s law), the effective food weight has been falling over time—from roughly 34 percent in the early 2000s to 28 percent after the 2016 basket revision—but remains substantially higher than in OECD economies.

The NBS constructs the CPI using a modified Laspeyres index with a five-year base-period update cycle. A consequential change occurred in 2016, when the NBS merged the previously separate urban and rural CPI baskets into a unified national basket. Before 2016, rural and urban price indices used different item lists and different outlet samples, creating discontinuities in province-level CPI series at the merge point. I begin the analysis in 2016 to work with a consistent basket throughout.

Within food, pork alone accounts for around 60 percent of total meat consumption in China. The NBS does not publish the exact weight of individual food items, but researchers who have backed out pork’s contribution by regressing headline CPI on sub-indices consistently place the pork weight at 2.5–4.5 percent for the 2016–2020 basket, with higher effective weights in provinces where per-capita pork consumption exceeds the national mean.

That official weight understates pork’s effective salience. Pork is purchased frequently: unlike durable categories whose prices households observe intermittently, pork is a daily staple for a large share of the population. Purchase frequency amplifies the informational weight that consumers assign to price changes relative to budget share (Cavallo, Cruces, and Perez-Truglia). The same logic applies with particular force in China, where wet markets are the primary retail channel for fresh protein and where household members

observe pork prices at every market visit—often daily in urban areas.

Historically, pork price cycles have been the single most important contributor to headline CPI volatility in China. The so-called “pig cycle” has generated four distinct peaks in pork retail prices since 2005: in 2007–2008, 2010–2011, 2015–2016, and the ASF-driven peak of 2019–2020. During each episode, year-on-year pork price inflation exceeded 50 percent for at least two consecutive months, and the pork sub-index accounted for 60–80 percent of the total variation in food CPI. The 2019 episode was the most extreme in the modern CPI record: the NBS meat and poultry sub-index reached 122 percent year-on-year in February 2020.

The baseline 28 percent food weight creates a strong prior that food-price movements are informative about aggregate inflation. When pork is expensive, households that weight recent experience heavily form upward-revised expectations even if non-food prices are stable. This channel is not irrational in a narrow accounting sense because food prices enter headline CPI. The issue is magnitude. The expectation response can exceed what food weights justify.

## **2.2 African Swine Fever as a Natural Experiment**

African Swine Fever (ASF) is a highly contagious viral disease affecting domestic and wild pigs with near-100 percent mortality in infected herds. It has no licensed vaccine, no cure, and no transmission pathway to humans. These features make ASF exceptionally difficult to contain once it enters a country’s swine population.

China confirmed its first domestic case in Shenyang, Liaoning province, in August 2018. Epidemiological tracing suggested that contaminated swill feed and long-distance live-pig transport were the initial vectors. Within two months, ASF had been confirmed in Heilongjiang, Henan, Jiangsu, and Zhejiang, spanning a 2,000-kilometer north-south corridor. By February 2019, every mainland province except Hainan had reported at least one outbreak.

The geographic pattern of spread reflected China’s live-pig trade network. Northeastern provinces (Liaoning, Heilongjiang, Jilin) are net exporters of pigs to consuming regions in the southeast (Guangdong, Fujian, Zhejiang). Long-distance truck transport moved infected animals across provincial borders before local outbreaks were detected.

The Ministry of Agriculture imposed interprovincial transport bans in September 2018, but enforcement was incomplete, and small-scale farmers—roughly 45 percent of national pork production—continued informal cross-border sales.

Between August 2018 and mid-2019, the official culling count exceeded 1.1 million pigs, but independent estimates from the FAO placed the total herd reduction at 30 to 55 percent of the pre-ASF national stock—roughly 120 to 210 million animals. Wholesale pork prices remained roughly flat through the first five months after the initial outbreak as frozen stocks buffered supply. Starting in March 2019, prices began a sustained increase. By October 2019, the NBS retail pork price index stood at approximately double its August 2018 level. The price increase was geographically uneven: provinces with larger pre-ASF herd densities (Sichuan, Henan, Hunan, Hubei) experienced the sharpest increases, while coastal provinces with cold-chain import infrastructure partially offset domestic supply losses with imports from the EU, Brazil, and the US.

The price shock reversed between 2020 and 2022. As surviving large-scale producers expanded capacity, the national herd rebuilt faster than expected; by mid-2021, hog inventories had returned to 95 percent of pre-ASF levels. Pork prices fell below pre-ASF levels in late 2021. This reversal is consequential for identification. Households who expected continued high inflation after observing the 2018–2020 meat-price surge were wrong about the direction, not just the magnitude, of subsequent price movements.

Two features of ASF matter for the identification strategy. The disease spread through biological transmission in pig populations, not as a consequence of local economic conditions or household demand. Cross-province variation in meat prices is therefore plausibly orthogonal to demand shocks that might independently affect inflation expectations. In addition, province-level treatment intensity was determined by pre-existing herd geography, centuries of agricultural specialisation, climate suitability, and feed-cost differentials, rather than by contemporary household characteristics. This pre-determination rules out the concern that provinces with systematically different expectation-formation processes selected into larger shocks.

## 2.3 Related Literature

### 2.3.1 Overreaction in expectations

The standard test for overreaction regresses forecast errors on prior forecast revisions (Coibion and Gorodnichenko). Under noisy information or sticky updating, the slope is positive: agents underreact. A negative slope indicates overreaction. Coibion and Gorodnichenko find positive slopes for US professional forecasters across most macroeconomic variables. Bordalo et al. revisit this test with a longer panel and document a sign switch: negative slopes at short horizons (overreaction), positive at long horizons (underreaction). The diagnostic expectations model of Bordalo, Gennaioli, and Shleifer generates this horizon dependence through a single parameter  $\theta$  governing the overweighting of the likelihood ratio.

A concern with the revision-error approach is that dispersed information can generate aggregate overreaction patterns even when individual agents update rationally (Angelos, Huo, and Sastry). My design partly sidesteps this problem: the treatment varies across provinces within a wave, so identification does not rely on aggregate time-series patterns. The design also avoids the mechanical correlation in the standard revision-error framework. The treatment is an NBS price statistic containing no household expectation data, and the dependent variable is constructed from the household's expectation and subsequent realised CPI, both measured independently of the treatment.

### 2.3.2 Salience and inflation expectations

Cavallo, Cruces, and Perez-Truglia show that US households weight frequently purchased goods more heavily in their inflation perceptions than official CPI weights imply. Perceived inflation tracks a frequency-weighted index roughly twice as closely as it tracks the official CPI. D'Acunto et al. extend this finding to European households: country-level variation in prices of frequently purchased goods predicts cross-country variation in inflation expectations after controlling for official CPI, with an asymmetry favouring price increases. Malmendier and Nagel show that personally experienced inflation shapes expectations decades later, and Kuchler and Zafar document an analogous pattern for housing.

These findings motivate the commodity-specific design. If salience operates through purchase frequency and personal exposure, meat and grain shocks should produce different expectation responses despite similar CPI weights. The placebo test exploits this asymmetry: grain shocks of comparable CPI weight do not produce the same forecast errors, ruling out mechanical pass-through.

### 2.3.3 China-specific evidence

Direct evidence on household inflation-expectation formation in China is scarce. Most existing work relies on aggregate diffusion indices from the PBoC Urban Depositor Survey, a quarterly survey of roughly 20,000 depositors in 50 cities (see, e.g., [Chen and Zha](#)). The aggregate nature of these data limits identification of cross-sectional expectation responses, and the survey samples depositors rather than the general population.

The CFPS micro data address both limitations. The CFPS is a nationally representative panel covering urban and rural households across 25 provinces, and province-level treatment variation allows me to test whether the expectation response differs across locations with different shock magnitudes. No prior work on Chinese inflation expectations has exploited the ASF episode as identifying variation. Existing ASF studies in economics focus on supply-chain disruption and trade effects; none examines the household belief response.

## 3 Conceptual Framework

This section derives a rational benchmark for the household forecast error and shows how diagnostic expectations generate overshooting. The model is deliberately parsimonious: its purpose is to discipline the empirical test, not to estimate structural parameters. Full derivations appear in [Appendix A](#).

### 3.1 Setup

A household in province  $p$  at survey wave  $t$  observes the cumulative meat-price change  $s_{pt}$  between waves  $t - 1$  and  $t$ . The household must forecast headline CPI inflation  $\pi_{p,t \rightarrow t+1}$  over the subsequent inter-wave period.



Meat enters headline CPI with weight  $\omega \approx 0.05$ . The meat-price shock has persistence  $\rho \in [0, 1]$ : a fraction  $\rho$  of the current shock carries forward into the next period's headline inflation. Formally, the headline CPI response to a meat shock of size  $s$  is

$$\mathbb{E}[\pi_{p,t \rightarrow t+1} \mid s_{pt}] = \bar{\pi} + \omega \rho s_{pt}, \quad (1)$$

where  $\bar{\pi}$  is unconditional expected inflation. The household does not know  $\rho$  with certainty but holds a prior belief  $\hat{\rho}$ .

I focus on persistence uncertainty rather than weight uncertainty because CPI weights are approximately stable across the sample period, so persistent disagreement about  $\omega$  would produce a constant expectation bias absorbed by fixed effects. Persistence uncertainty, by contrast, interacts with the size of the shock and generates a coefficient on  $s_{pt}$  in the forecast-error regression.

### 3.2 Rational Benchmark

Under rational expectations, the household forecasts  $\mathbb{E}_t^R[\pi_{p,t \rightarrow t+1}] = \bar{\pi} + \omega \hat{\rho} s_{pt}$ , where  $\hat{\rho}$  is the household's best estimate of persistence. The forecast error is

$$\text{FER}_{pt}^R = \pi_{p,t \rightarrow t+1} - \mathbb{E}_t^R[\pi_{p,t \rightarrow t+1}] = \omega(\rho - \hat{\rho}) s_{pt} + \eta_{pt}, \quad (2)$$

where  $\eta_{pt}$  collects shocks to headline CPI that are orthogonal to the meat signal. If the household overestimates persistence ( $\hat{\rho} > \rho$ ), it expects more meat-driven inflation than actually occurs, and the forecast error is negative in provinces with positive shocks. If the household underestimates persistence, the forecast error is positive in high-shock provinces.

The coefficient on  $s_{pt}$  in a regression of the forecast error on the meat shock is therefore bounded:

$$\beta_3^R = \omega(\rho - \hat{\rho}). \quad (3)$$

Even in the worst case—where households believe the shock is permanent ( $\hat{\rho} = 1$ ) but it is fully transitory ( $\rho = 0$ )—the rational forecast-error coefficient is  $\beta_3^R = -\omega \approx -0.05$ . Under any intermediate belief,  $|\beta_3^R| < \omega$ . The rational benchmark therefore predicts

$|\beta_3| \leq 0.05$  per unit of the meat shock.

This bound is the central object of the empirical test. The number 0.05 is not a free parameter—it is the CPI weight on meat, determined by the NBS basket construction. Any rational household, regardless of its belief about persistence, generates a forecast-error coefficient no larger than the mechanical weight of meat in the index, because persistence governs the *duration* of the meat contribution, not its *size per period*.

### 3.3 Time Aggregation

The one-period model generates a bound  $|\beta_3^R| \leq \omega$  per unit of a single-period shock. The empirical treatment is a cumulative inter-wave shock  $S_{pt} = \sum_{m=1}^M s_{pm}$  spanning roughly 24 months. This subsection shows that the bound extends to the time-aggregated object.

Under the AR(1) specification, the headline CPI contribution of a single monthly meat shock  $s_{pm}$  at horizon  $h$  is  $\omega \rho^h s_{pm}$ . The cumulative forward headline CPI response to the sequence of monthly shocks is

$$\sum_{m=1}^M \sum_{h=1}^H \omega \rho^h s_{pm} = \omega \frac{\rho(1 - \rho^H)}{1 - \rho} S_{pt}, \quad (4)$$

where  $H$  is the forward horizon in months and I assume for simplicity that each monthly shock contributes equally to the cumulative treatment  $S_{pt}$ . A rational household that knows  $\rho$  forecasts this quantity exactly; a household that believes  $\hat{\rho}$  forecasts the same expression with  $\hat{\rho}$  in place of  $\rho$ . The forecast-error coefficient on  $S_{pt}$  is therefore

$$\beta_3^R = \omega \left[ \frac{\rho(1 - \rho^H)}{1 - \rho} - \frac{\hat{\rho}(1 - \hat{\rho}^H)}{1 - \hat{\rho}} \right]. \quad (5)$$

The term in brackets is a difference of two bounded functions of persistence. For any  $\rho, \hat{\rho} \in [0, 1]$  and finite  $H$ , the absolute value of the bracket is bounded above by  $H$  (the maximum number of contributing periods). The rational forecast-error coefficient per unit of the cumulative shock therefore satisfies  $|\beta_3^R| \leq \omega \cdot H/M$ . With  $M \approx H \approx 24$  (both the treatment window and the forward window span roughly two years), the bound simplifies to  $|\beta_3^R| \leq \omega \approx 0.05$ , exactly as in the one-period case.

The bound is weakest when  $M$  is small relative to  $H$ , so that a concentrated shock

has a long forward horizon over which persistence beliefs matter. Even in the extreme case  $M = 1$  (a single monthly shock) with  $H = 24$  and  $\hat{\rho} = 1$ , the rational bound is  $|\beta_3^R| \leq 24\omega \approx 1.2$ , still far below the estimated coefficient of  $|\hat{\beta}_3| \approx 7.4$ . The time-aggregation mismatch therefore cannot close the gap between the rational bound and the data. Full derivations appear in Appendix A.

### 3.4 Diagnostic Extension

Under diagnostic expectations (Bordalo, Gennaioli, and Shleifer), the household overweights the state that recent signals make more representative. A large positive meat-price shock makes high-inflation states more salient, so the household inflates its forecast by a factor  $\theta > 0$  beyond the rational update:

$$\mathbb{E}_t^\theta[\pi_{p,t \rightarrow t+1}] = \bar{\pi} + \omega \hat{\rho} s_{pt} + \theta \omega s_{pt}. \quad (6)$$

The additional term  $\theta \omega s_{pt}$  arises because the household tilts its subjective probability distribution toward states with high likelihood ratios. When  $\theta = 0$ , the household reverts to the rational benchmark; when  $\theta > 0$ , it places too much probability mass on high-inflation outcomes after observing a positive meat-price shock.

The diagnostic forecast error is

$$\text{FE}_{pt}^\theta = \omega(\rho - \hat{\rho}) s_{pt} - \theta \omega s_{pt} + \eta_{pt}, \quad (7)$$

so the regression coefficient becomes  $\beta_3^\theta = \omega(\rho - \hat{\rho} - \theta)$ . With  $\theta > 0$ , the forecast error is more negative than under rationality for any belief  $\hat{\rho}$  and any true persistence  $\rho$ .

The forecast-error regression is a sharper test than the expectation regression. The expectation coefficient  $\beta_1$  depends on  $\omega(\hat{\rho} + \theta)$ , so distinguishing the rational component from the diagnostic component requires an independent estimate of  $\hat{\rho}$ , which is not directly observed. The forecast-error coefficient  $\beta_3$ , by contrast, is bounded by  $\omega$  under rationality regardless of  $\hat{\rho}$ . Any estimate of  $|\hat{\beta}_3|$  exceeding 0.05 rejects rationality without requiring knowledge of the household's belief about persistence.

### 3.5 Testable Implications

The framework generates three predictions that the data can discriminate.

*Prediction 1: Bounded forecast errors under rationality.* Under rationality, the forecast-error coefficient satisfies  $|\beta_3| \leq \omega \approx 0.05$ . If the estimated  $\hat{\beta}_3$  exceeds this bound in absolute value, rationality is rejected regardless of the household's belief about persistence. The bound is conservative: it assumes the worst-case belief ( $\hat{\rho} = 1$  when  $\rho = 0$ ); under any more accurate belief, the rational coefficient is smaller. In the data, the estimated coefficient is  $\hat{\beta}_3 = -7.61$  under the preferred specification (province plus region-by-wave fixed effects). Because the CFPS expectation is ordinal, the exact magnitude of  $\hat{\beta}_3$  depends on the coding of the ordinal variable. The test does not rely on this magnitude, however. The *sign* of  $\hat{\beta}_3 < 0$  rejects rationality under any monotone rescaling (Appendix E), and the pass-through coefficient  $\hat{\beta}_2 = -5.25$ , which involves no ordinal variable, independently exceeds the time-aggregated rational bound of 1.20 by a factor of four.

*Prediction 2: Excess expectation response.* Under diagnostic expectations the expectation response exceeds the rational update:  $\hat{\beta}_1 > \omega \hat{\rho}$ . The expectation result complements the forecast-error test by showing *where* the excess forecast error comes from: through the numerator (expectations are too high) rather than the denominator (realised CPI is too low). In the preferred specification,  $\hat{\beta}_1 = 0.119$  ( $p = 0.019$ ), which is positive and statistically significant at the five-percent level.

*Prediction 3: Commodity asymmetry.* A household that overweights salient signals should respond more to meat shocks (high purchase frequency, high media coverage) than to grain shocks of comparable CPI weight. The diagnostic model predicts that  $\theta$  is larger for meat than for grain because the purchase-frequency channel amplifies the representativeness of meat-price signals. Grain prices in China are stabilised by government procurement and strategic reserves, so they vary less and attract less household attention. The placebo test in Section 7 confirms this asymmetry: grain shocks produce near-zero forecast errors.

### 3.6 Mapping to Empirical Equations

The three predictions map directly to the three-equation system estimated in Section 5. Equation 1 regresses household expectations on the meat shock and tests whether  $\hat{\beta}_1 > 0$ . Equation 2 regresses forward realised headline CPI on the meat shock and estimates the actual pass-through  $\omega \rho$ . Equation 3 regresses the forecast error (realised CPI minus expectation) on the meat shock and tests whether  $|\hat{\beta}_3| > \omega$ .

The empirical strategy uses province fixed effects combined with region-by-wave fixed effects in the preferred specification. These fixed effects absorb time-invariant province characteristics and any time-varying shocks common to provinces within a region. The identifying variation is the within-province deviation of the meat shock from the region-by-wave mean. The model predicts that the forecast-error bound  $|\beta_3| \leq 0.05$  holds even in this demanding specification because the bound derives from the CPI weight, not from the level of the shock.

Section 6.4 compares the estimated  $\hat{\beta}_3$  against the rational bound calibrated to the actual persistence of meat-price shocks in the data.

## 4 Data

### 4.1 China Family Panel Studies

The China Family Panel Studies (CFPS) is a nationally representative biennial household panel survey administered by the Institute of Social Science Survey (ISSS) at Peking University ([Institute of Social Science Survey, Peking University](#)). The baseline wave in 2010 drew a stratified, multi-stage probability sample of approximately 16,000 households in 25 mainland provinces, covering roughly 95 percent of the national population. The survey has been fielded in 2010, 2012, 2014, 2016, 2018, 2020, and 2022. I use the four waves from 2016 to 2022, which overlap with the availability of province-level meat CPI data from the NBS. For auxiliary cross-sectional checks in companion specifications, I draw CHFS source files from the China Household Finance Survey and Research Center at Southwestern University of Finance and Economics ([China Household Finance Survey and Research Center, Southwestern University of Finance and Economics](#)).

Response rates are high by international panel standards. The baseline household response rate was approximately 81 percent, with re-contact rates between 75 and 85 percent in subsequent waves. Attrition is non-random—younger, more educated, and urban-resident households are more likely to exit—and I include age, education, and urban indicators as controls where available.

The inflation expectation variable is a three-point ordinal question (item qp201) that asks respondents: “Compared with the current period, do you think the price level in the coming year will rise, stay the same, or fall?” I code the response as  $\mu_{it} \in \{-1, 0, 1\}$ , where 1 indicates expected price increases and  $-1$  indicates expected decreases. The question wording refers to “price level” (物价水平) rather than the CPI specifically, which is consistent with the salience channel: if respondents map “price level” to the prices of goods they buy frequently, their responses will be disproportionately influenced by salient price changes.

The coarseness of the three-point scale attenuates the estimated coefficients: respondents who revise their beliefs upward by different amounts all register as  $+1$ . This attenuation bias works against finding overreaction, so any detected overreaction is a lower bound.

The estimation sample contains 90,133 household-wave observations across 31 provinces and four waves. The forecast-error sample is smaller (69,529 observations) because the 2022 wave lacks forward realised headline CPI at the time of data construction. Demographic controls—age, a high-education indicator, and an urban-residence indicator—are available for a subsample of approximately 22,100 observations.

## 4.2 Province-Level CPI Data

The National Bureau of Statistics (NBS) publishes monthly consumer price indices at the province level for major sub-categories. I use the meat and poultry sub-index, the grain sub-index, and the headline CPI index. Province-level meat CPI is available from January 2016 onward; grain and headline CPI are available from January 2012.

Geographic coverage spans all 31 mainland provinces. Two potential measurement concerns arise. The outlet sample may not fully represent rural areas, where wet markets are the dominant channel for fresh protein. If rural meat prices move differently from the

sampled outlets, the NBS meat sub-index may understate the price variation experienced by rural CFPS households, attenuating the estimated coefficients toward zero. The transition from separate urban and rural CPI baskets to a unified basket in 2016 may also create a level shift in some province-level series. I avoid this problem by beginning the analysis in 2016 and using cumulative log changes, which difference out level shifts.

The NBS reports these sub-indices in *previous month* = 100 form: a value of 105 means the price level rose 5 percent relative to the preceding month. I convert each monthly observation to a log change by dividing by 100 and taking the natural logarithm; because the base is the previous month, this yields the exact month-on-month log growth rate. I then sum these monthly log changes over the inter-wave period to obtain the cumulative treatment.

### 4.3 Treatment Construction

The treatment variable is the cumulative log change in provincial meat CPI between two consecutive CFPS waves:

$$\text{MeatShock}_{p,t-1 \rightarrow t} = \sum_{m \in \mathcal{M}(t-1,t)} \log\left(\frac{\text{MeatCPI}_{pm}}{100}\right), \quad (8)$$

where  $\text{MeatCPI}_{pm}$  is the NBS meat and poultry sub-index for province  $p$  in month  $m$ , and  $\mathcal{M}(t-1,t)$  is the set of months between the midpoints of two consecutive CFPS fieldwork periods. The grain shock is constructed identically from the grain sub-index.

I define the inter-wave period using the midpoint of each wave’s fieldwork window. The 2016 wave was fielded primarily between June and December 2016; the 2018 wave between June and November 2018; the 2020 wave between July 2020 and January 2021 (delayed by COVID-19); and the 2022 wave between June and December 2022. This midpoint convention ensures that the treatment captures the price changes that the typical respondent experienced between consecutive interview dates.

The identifying variation operates at the province×wave level. The sample contains 93 province×wave cells across 31 provinces and three to four waves, depending on forward CPI availability. All standard errors are clustered at the province level (31 clusters), and inference uses the wild cluster bootstrap to account for the small number of clusters.

## 4.4 Descriptive Statistics

**Table 1:** Descriptive Statistics

| Variable                             | $N$    | Mean  | SD    | Min   | Median | Max   |
|--------------------------------------|--------|-------|-------|-------|--------|-------|
| Inflation expectation ( $\mu_{it}$ ) | 90,133 | -0.32 | 0.81  | -1.00 | -1.00  | 1.00  |
| Forecast error ( $FE_{it}$ )         | 69,533 | 4.67  | 1.88  | -0.21 | 5.07   | 8.14  |
| Meat shock ( $s_{pt}$ )              | 90,133 | 0.05  | 0.28  | -0.40 | -0.01  | 0.57  |
| Grain shock                          | 90,133 | 0.04  | 0.02  | -0.07 | 0.03   | 0.12  |
| Realized headline CPI (% , ann.)     | 69,533 | 4.03  | 0.96  | 1.79  | 3.86   | 6.14  |
| Age                                  | 22,320 | 40.19 | 14.27 | 18.00 | 39.00  | 70.00 |
| High education (indicator)           | 90,133 | 0.30  | 0.46  | 0.00  | 0.00   | 1.00  |
| Urban (indicator)                    | 87,414 | 0.67  | 0.47  | 0.00  | 1.00   | 1.00  |

*Note.* Sample restricted to observations with non-missing province code, inflation expectation, and meat shock.

Inflation expectation coded as  $\{-1, 0, 1\}$  (prices fall, stable, rise).

Forecast error = realized headline CPI (%) – ordinal expectation ( $\{-1, 0, 1\}$ ).

Demographics available for a subsample;  $N$  varies across rows.

Table 1 reports summary statistics for the estimation sample.

The mean inflation expectation is  $-0.32$  on the  $[-1, 1]$  scale, indicating that the majority of respondents expect prices to stay the same or fall. This finding is consistent with two features of the data. The sample period includes the post-ASF price reversal of 2021–2022, during which headline CPI inflation fell below 1 percent. The  $\{-1, 0, 1\}$  coding also treats “stay the same” as the midpoint.

The mean forecast error is 4.67 percentage points. The positive sign reflects the mechanical asymmetry of the ordinal coding: when respondents report expectations of zero or below but realised CPI inflation is positive, the forecast error is positive. The regression analysis controls for this level effect through fixed effects; the coefficient of interest captures the *differential* forecast error in high-shock versus low-shock provinces, not the mean level.

The mean meat shock is 0.05 log points, with a standard deviation of 0.28. The near-zero mean reflects offsetting waves: the 2018–2020 pair captures the ASF peak (large positive shocks), while the 2020–2022 pair captures the price reversal (large negative shocks). A one-standard-deviation shock of 0.28 log points corresponds to a cumulative meat-price increase of roughly 32 percent ( $e^{0.28} - 1 \approx 0.32$ ) over the inter-wave period.

Forward realised headline CPI averages 4.03 percent (annualised), with moderate cross-province variation driven by the elevated food prices during the 2018–2020 inter-



wave period.

## 5 Empirical Strategy

The strategy exploits province-level meat-price variation as an external salience treatment. The treatment is measured by the National Bureau of Statistics and merged from outside the CFPS survey, so the household’s current expectation never appears on both sides of any regression. The primary test is a reduced-form regression of forecast errors on the meat shock. Two decomposition equations identify the channels through which the forecast error arises.

### 5.1 Treatment Variable

The treatment variable is the cumulative log change in provincial meat CPI between two consecutive CFPS waves, defined in equation (8). Three properties matter for identification. The variable is external to the survey, containing no information derived from respondents. It is cumulative: a household surveyed in 2020 experienced the full path of meat-price movements since the 2018 wave. And it varies at the province level, which allows me to absorb region-by-wave fixed effects while retaining within-region identifying variation.

The cumulative measure is the natural choice given the CFPS survey design. CFPS is fielded every two years, with interviews spanning several months within each wave. Households report expectations once per wave, and those reports reflect whatever price information they accumulated since the prior interview. The cumulative log change between wave midpoints therefore captures the total price signal that a household plausibly processed.

Two measurement considerations deserve attention. The NBS meat CPI index combines pork, beef, mutton, and poultry, with pork receiving the dominant weight. During the ASF episode, pork drove the bulk of the meat CPI movement, so the index tracks the pork shock with high fidelity during the identification window. The cumulative measure also aggregates monthly observations equally. If households weight recent months more heavily, the treatment variable understates effective salience, attenuating the estimated

coefficient toward zero.

The 2018–2019 ASF episode provides the sharpest source of variation. Provincial herd losses varied because culling policy, transport restrictions, and initial herd density differed across provinces. This heterogeneity generates cross-sectional dispersion in the treatment that is plausibly orthogonal to province-level demand conditions.

## 5.2 Three-Equation System

Primary test: forecast errors (Equation 3). The reduced-form implication of overreaction is that households in high-shock provinces subsequently overpredict inflation:

$$\text{FE}_{ipt} = \alpha_p + \beta_3 \text{MeatShock}_{p,t-1 \rightarrow t} + \gamma' X_{ipt} + \delta_{r(p) \times t} + \varepsilon_{ipt}, \quad (9)$$

where  $\alpha_p$  is a province fixed effect,  $\delta_{r(p) \times t}$  is a region-by-wave fixed effect, and  $\text{FE}_{ipt} \equiv \pi_{p,t \rightarrow t+1}^{\text{headline}} - \tilde{\mu}_{ipt}$  is realised provincial headline inflation over the next inter-wave period minus the household’s directional ordinal expectation ( $\mu_{ipt} \in \{-1, 0, 1\}$ ). A negative forecast error means that the household expected more inflation than materialised. Under overreaction,  $\beta_3 < 0$ : high-shock provinces generate more negative forecast errors.

The forecast-error equation occupies the primary position because it requires the fewest auxiliary assumptions. The sign of  $\beta_3$  alone tests overreaction. If households responded rationally, their expectations would adjust by no more than the shock’s actual headline CPI weight, and forecast errors would be bounded by  $\omega \approx 0.05$  regardless of the household’s belief about shock persistence (Section 3). Any  $|\hat{\beta}_3|$  exceeding this bound rejects rationality without requiring knowledge of the conversion factor between ordinal expectations and CPI units.

The preferred specification includes province fixed effects together with region-by-wave fixed effects. Province fixed effects absorb any time-invariant provincial characteristic (income level, food culture, baseline inflation expectations); region-by-wave fixed effects absorb time-varying shocks common to provinces within a region. The identifying variation is the within-province deviation of the meat shock from the region-by-wave mean. This double-differencing design asks whether a province that receives a larger-than-regional-average meat shock in a given wave produces a more negative forecast

error than the same province does at other times.

Decomposition: expectations (Equation 1).

$$\mu_{ipt} = \alpha_p + \beta_1 \text{MeatShock}_{p,t-1 \rightarrow t} + \gamma' X_{ipt} + \delta_{r(p) \times t} + \varepsilon_{ipt}, \quad (10)$$

where  $\mu_{ipt} \in \{-1, 0, 1\}$  is the ordinal inflation expectation. A positive  $\hat{\beta}_1$  indicates that higher meat-price shocks shift expectations upward.

Equation 1 identifies whether the shock moves beliefs. If households are inattentive or if meat prices lack salience,  $\beta_1$  would be zero. Because the dependent variable is ordinal, the coefficient measures shifts in the probability of reporting “up” versus “same” or “down,” not a cardinal change in expected inflation. The ordinal scale limits what  $\beta_1$  can tell us about the magnitude of the expectation response in CPI units, which is precisely why the forecast-error equation is the primary test.

Decomposition: pass-through (Equation 2).

$$\pi_{p,t \rightarrow t+1}^{\text{headline}} = \alpha + \beta_2 \text{MeatShock}_{p,t-1 \rightarrow t} + \delta_{r(p) \times t} + u_{pt}, \quad (11)$$

where  $\pi_{p,t \rightarrow t+1}^{\text{headline}}$  is realised province-level headline CPI inflation over the subsequent inter-wave period. This regression runs at the province-wave level (93 observations). If  $\beta_2 \leq 0$ , the shock does not raise future headline inflation, closing the second half of the overreaction logic.

The pass-through equation pins down what a rational household should have expected. A zero or negative  $\beta_2$  means the shock reverses. Provinces that received large meat-price increases subsequently saw lower, not higher, headline inflation. The reversal follows ASF supply-shock dynamics. As herds recovered through 2021–2022, pork supply expanded and meat prices fell, dragging down headline CPI in the provinces that experienced the sharpest earlier increases. The 93-observation sample size motivates the wild cluster bootstrap described in Section 5.4.

### 5.2.1 Decomposition Logic and the Overreaction Test

The three coefficients are linked by an approximate decomposition:  $\beta_3 \approx \beta_2 - k \beta_1$ , where  $k$  converts the ordinal expectation into implied inflation units. This identity clarifies the

overreaction test.

The forecast error is realised inflation minus expected inflation. Regressing realised inflation on the meat shock gives  $\beta_2$ . Regressing expected inflation on the meat shock gives  $k\beta_1$ . If expectations respond ( $\beta_1 > 0$ ) but realised CPI does not ( $\beta_2 \leq 0$ ), the forecast error is necessarily negative because the household raised its expectation in response to a signal that did not raise the object it was forecasting.

The sign test  $\beta_3 < 0$  does not require knowledge of  $k$ . The CFPS expectation question is ordinal, so the mapping from  $\mu \in \{-1, 0, 1\}$  to CPI units is unknown. But the forecast error is constructed in CPI units, so  $\beta_3$  is directly interpretable. Whatever  $k$  is, if the forecast error is systematically more negative in high-shock provinces, households in those provinces overpredicted inflation.

The decomposition also provides a scaling-free magnitude test. Because  $\beta_3 \approx \beta_2 - k\beta_1$ , setting  $k = 0$  (ignoring the ordinal expectation entirely) yields  $\beta_3 = \beta_2$ . If  $|\hat{\beta}_2|$  exceeds the rational bound, the conclusion holds without assigning any cardinal meaning to the ordinal codes. Under the time-aggregated bound (Section 3.3),  $|\beta_3^R| \leq 24\omega \approx 1.20$ . Finding  $|\hat{\beta}_2| = 5.25$  in the preferred specification exceeds this bound by a factor of four, independent of any scaling assumption.

### 5.3 Identification Assumptions

The strategy requires two conditions.

The first condition is relevance: the meat-price shock must predict household forecast errors. Pork is widely consumed, purchased frequently, and priced visibly; ASF generated extensive media coverage that was difficult to miss even for inattentive consumers. Three features support relevance: pork accounts for roughly 60 percent of China’s meat consumption and is purchased several times per week; the ASF episode received extensive state and social media coverage; and prices at wet markets are posted on visible boards and updated daily.

The second condition is conditional exogeneity: the meat-price shock must be uncorrelated with province-level determinants of forecast errors other than through salience, conditional on fixed effects and controls. Region-by-wave fixed effects absorb regional business cycles, monetary policy transmission, and nationwide shocks hitting all provinces

within a region simultaneously. The remaining identifying variation is within-region cross-province differences in meat-price shocks over time.

The main threat is a province-specific demand shock that simultaneously raises meat prices and expectations through a real-income channel. Two features mitigate this concern. ASF provides a biologically determined supply shock: the disease entered China through contaminated feed imports and spread along pig transportation corridors, with provincial infection severity determined by initial herd density and proximity to early outbreak sites. The commodity-placebo test in Section 7 provides further discipline: grain shocks of comparable magnitude do not produce the same forecast-error pattern. If local demand were driving both prices and expectations, grain prices should respond to the same demand shock and produce a similar expectation effect.

A second concern is that province-level meat-price variation might proxy for local food-price inflation more broadly. Region-by-wave fixed effects partially address this by absorbing region-level food-price trends. The within-region variation in meat prices is largely idiosyncratic to the ASF episode and does not coincide with similar dispersion in non-meat food categories.

The design also requires stable unit treatment: household  $i$ 's forecast error depends on the meat shock in province  $p$  where the household resides, not on meat shocks in other provinces. Region-by-wave fixed effects absorb any national-level or region-level price signal, so the identifying variation reflects local price conditions. The CFPS expectation question asks about general price changes, not pork prices specifically; households that follow national pork-price news would report higher expectations in all provinces, which would be absorbed by the wave fixed effect.

What the fixed effects absorb and what they leave. Province fixed effects absorb permanent provincial characteristics: income, food culture, media environment, baseline inflation expectations. Region-by-wave fixed effects absorb any time-varying shock common to provinces within a region, including aggregate monetary policy, regional business cycles, and region-specific food supply disruptions. Together, the two sets of fixed effects leave only within-province, within-region-wave variation in meat-price shocks as the identifying source. Robustness checks in Section 8 show that results are stable across alternative fixed-effect structures, including region-by-wave alone and province-plus-wave.

If a province experienced an unobserved demand boom that raised both meat prices and expectations, this would bias  $\beta_3$  toward zero rather than away from zero. The direction of potential omitted-variable bias therefore works against the overreaction finding.

## 5.4 Inference

With 31 province clusters, asymptotic cluster-robust standard errors can over-reject (Cameron, Gelbach, and Miller). Simulation evidence suggests that 30–50 clusters is a borderline case where asymptotic approximations may or may not perform well.

I report wild cluster bootstrap  $p$ -values throughout, using the Webb six-point weight distribution with 999 replications. The six-point distribution is preferred over the Rademacher two-point distribution because it provides better size control for cluster counts between 10 and 50, precisely the range that applies here. The bootstrap is implemented under the null hypothesis of zero treatment effect (restricted residuals).

Point estimates and conventional cluster-robust standard errors are reported alongside bootstrap  $p$ -values. All significance statements in the text refer to bootstrap  $p$ -values unless otherwise noted.

## 6 Main Results

The primary finding is that province-level meat-price shocks generate large, negative household forecast errors that lie well above the rational benchmark in absolute value. I present the forecast-error result first (Section 6.1), then decompose it into the expectation channel (Section 6.2) and the pass-through channel (Section 6.3). Section 6.4 compares the estimated coefficient against the rational bound. Section 6.5 visualises the treatment variation.

### 6.1 Meat Shocks Generate Systematic Forecast Errors

Table 2 reports estimates of equation (9). The dependent variable is realised province-level headline CPI inflation minus the household’s inflation expectation. A negative value indicates overprediction: the household expected more inflation than materialised.

**Table 2:** Meat Price Shock and Household Forecast Errors

|                  | (1)                  | (2)               | (3)                  | (4)                  | (5)                             | (6)                         |
|------------------|----------------------|-------------------|----------------------|----------------------|---------------------------------|-----------------------------|
|                  | Bivariate            | Demographics      | Province<br>+ wave   | Region<br>× wave     | Region × wave<br>+ demographics | Province +<br>region × wave |
| Meat shock       | −1.960***<br>(0.259) | −2.860<br>(5.783) | −5.694***<br>(1.197) | −6.291***<br>(1.129) | −6.628<br>(5.373)               | −7.610***<br>(1.124)        |
| Demographics     |                      | Yes               |                      |                      | Yes                             |                             |
| Province FE      |                      |                   | Yes                  |                      |                                 | Yes                         |
| Wave FE          |                      |                   | Yes                  |                      |                                 |                             |
| Region × wave FE |                      |                   |                      | Yes                  | Yes                             | Yes                         |
| Observations     | 69,529               | 22,118            | 69,529               | 69,529               | 22,118                          | 69,529                      |
| $R^2$            | 0.215                | 0.023             | 0.828                | 0.801                | 0.154                           | 0.887                       |

*Note.* The dependent variable is the household forecast error: realized province-level CPI inflation (annual %) minus the Carlson-Parkin quantified expected inflation ( $\delta = 0.5$  pp). Standard errors clustered at the province level in parentheses. Column (6) is the preferred specification.

\* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

The coefficient on the meat shock is negative in every column, and statistically significant at the one-percent level in the four specifications that use the full sample (columns 1, 3, 4, and 6). The bivariate estimate (column 1) is  $\hat{\beta}_3 = -1.97$ . Adding province-plus-wave fixed effects (column 3) raises the absolute value to 5.69; region-by-wave fixed effects alone (column 4) give 6.29. The preferred specification adds province fixed effects on top of region-by-wave fixed effects (column 6), yielding  $\hat{\beta}_3 = -7.61$  (bootstrap  $p = 0.005$ ).

The monotone increase in  $|\hat{\beta}_3|$  as more fixed effects enter is the opposite of what one would expect if the result were driven by omitted confounds. Absorbing province-level permanent heterogeneity and region-by-wave shocks strengthens, rather than attenuates, the finding.

Columns 2 and 5 add household demographics. Both are imprecise because the demographic subsample drops to 22,121 observations. The point estimates retain the correct sign; the precision loss is mechanical given a 68 percent reduction in sample size with the same number of clusters.

The preferred estimate has a clear economic interpretation. Consider two provinces in the same region observed in the same wave; one experienced a cumulative meat CPI increase of 0.5 log points (about 65 percent), the other 0.3 log points (about 35 percent). The difference in treatment is 0.2 log points. The predicted difference in forecast errors is  $-7.61 \times 0.2 = -1.52$ : households in the higher-shock province overpredict inflation by 1.52 percentage points more, roughly three-quarters of the average headline CPI rate

during the sample period.

## 6.2 Decomposition: Expectations

**Table 3:** Meat Price Shock and Household Inflation Expectations

|                  | (1)                | (2)                 | (3)                | (4)              | (5)                             | (6)                         |
|------------------|--------------------|---------------------|--------------------|------------------|---------------------------------|-----------------------------|
|                  | Bivariate          | Demographics        | Province<br>+ wave | Region<br>× wave | Region × wave<br>+ demographics | Province +<br>region × wave |
| Meat shock       | 0.028**<br>(0.012) | 2.492***<br>(0.812) | 0.090*<br>(0.053)  | 0.242<br>(0.174) | 2.636***<br>(0.806)             | 0.119**<br>(0.050)          |
| Demographics     |                    | Yes                 |                    |                  | Yes                             |                             |
| Province FE      |                    |                     | Yes                |                  |                                 | Yes                         |
| Wave FE          |                    |                     | Yes                |                  |                                 |                             |
| Region × wave FE |                    |                     |                    | Yes              | Yes                             | Yes                         |
| Observations     | 90,133             | 22,121              | 90,133             | 90,133           | 22,121                          | 90,133                      |
| $R^2$            | 0.000              | 0.038               | 0.011              | 0.002            | 0.042                           | 0.012                       |

*Note.* The dependent variable is the household’s ordinal inflation expectation ( $\mu_{it} \in \{-1, 0, 1\}$ ). Standard errors clustered at the province level in parentheses. Column (6) is the preferred specification.

\* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 3 asks whether meat-price shocks shift expectations (equation (10)). The coefficient is positive in all six columns.

The bivariate estimate (column 1) is  $\hat{\beta}_1 = 0.028$  ( $p = 0.02$ ): a one-unit increase in the cumulative meat log-change shifts the ordinal expectation upward by 0.028 units. Under province-plus-wave fixed effects (column 3), the coefficient rises to 0.090 ( $p = 0.095$ ). Region-by-wave fixed effects alone (column 4) yield a larger point estimate of 0.242 but with wider confidence intervals ( $p = 0.225$ ).

The preferred specification (column 6) adds province fixed effects to region-by-wave fixed effects:  $\hat{\beta}_1 = 0.119$  ( $p = 0.019$ ). The province fixed effects absorb permanent cross-province differences in the share of respondents reporting “up,” improving precision relative to the region-by-wave-only specification. The expectation channel is now statistically significant at the five-percent level, confirming that meat-price shocks move household beliefs, not only forecast errors.

The expectation result complements the forecast-error test. By itself,  $\hat{\beta}_3 < 0$  could arise either because households raised expectations too much or because realised CPI fell. The positive  $\hat{\beta}_1$  confirms that the numerator of the forecast error (expectations) moved in the wrong direction, not only the denominator (realised CPI).



### 6.3 Decomposition: Pass-Through

**Table 4:** Meat Price Shock and Realized Headline CPI

|                  | (1)                  | (2)                  | (3)                | (4)                         |
|------------------|----------------------|----------------------|--------------------|-----------------------------|
|                  | Bivariate            | Province<br>+ wave   | Region<br>× wave   | Province +<br>region × wave |
| Meat shock       | −1.812***<br>(0.232) | −4.703***<br>(1.646) | −3.519*<br>(1.954) | −5.249***<br>(1.817)        |
| Province FE      |                      | Yes                  |                    | Yes                         |
| Wave FE          |                      | Yes                  |                    |                             |
| Region × wave FE |                      |                      | Yes                | Yes                         |
| Observations     | 93                   | 93                   | 93                 | 93                          |
| $R^2$            | 0.192                | 0.704                | 0.589              | 0.738                       |

*Note.* Unit of observation is province × wave.

Standard errors clustered at the province level.

Dependent variable is annualized realized headline CPI inflation in the province over the subsequent inter-wave period.

\* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 4 estimates equation (11) at the province×wave level (93 observations). The point estimate is negative in all columns: provinces with larger meat-price shocks experienced lower, not higher, forward headline inflation.

The bivariate estimate is  $\hat{\beta}_2 = -1.81$  ( $p < 0.001$ ). Province-plus-wave fixed effects (column 2) yield  $-4.70$  ( $p = 0.008$ ); the preferred province-plus-region-by-wave specification (column 4) yields  $\hat{\beta}_2 = -5.25$  ( $p = 0.007$ ).

The negative  $\beta_2$  reflects the ASF supply-shock reversal. The ASF shock contracted pork supply in 2018–2019, raising meat prices sharply. As herds recovered through 2020–2022, the same provinces saw the sharpest price reversals: meat CPI fell fastest where it had risen most. Because meat carries a non-trivial weight in headline CPI (roughly 3–6 percent), the reversal dragged down headline CPI in high-shock provinces.

This reversal is central to the overreaction interpretation. Households in high-shock provinces observed large meat-price increases and revised their inflation expectations upward. But the price signal was temporary: future headline inflation in their province was not higher but lower. The household overreacted in the sense that it treated a transitory shock as informative about future inflation when the shock predicted the opposite.

## 6.4 Comparison with the Rational Benchmark

**Table 5:** Calibration: Estimated vs. Rational Forecast-Error Coefficient

| Persistence scenario      | $\rho$ | Rational bound ( $\beta_3^R$ ) |            | $\hat{\beta}_3$ | Ratio    |
|---------------------------|--------|--------------------------------|------------|-----------------|----------|
|                           |        | Correct belief                 | Worst case |                 |          |
| Transitory ( $\rho = 0$ ) | 0.00   | 0                              | −0.050     | −7.610          | 152×     |
| Estimated $\hat{\rho}$    | 0.48   | 0                              | −0.026     | −7.610          | 293×     |
| Moderate ( $\rho = 0.5$ ) | 0.50   | 0                              | −0.025     | −7.610          | 304×     |
| Permanent ( $\rho = 1$ )  | 1.00   | 0                              | 0          | −7.610          | $\infty$ |

*Note.*  $\omega = 0.05$  (meat weight in headline CPI).  $\hat{\rho} = 0.480$  estimated from AR(1) on monthly province-level meat CPI with province fixed effects.

Rational bound under correct belief:  $\beta_3^R = 0$  (household knows true  $\rho$ ).

Worst case: household believes shock is permanent ( $\hat{\rho} = 1$ ), giving  $\beta_3^R = -\omega(1 - \rho)$ .

$\hat{\beta}_3 = -7.610$  from the preferred specification (province + region  $\times$  wave FE, Table 2, column 6).

The ordinal expectation scale inflates the exact ratio; the point estimate should be read as an order-of-magnitude statement.

Table 5 compares the estimated  $\hat{\beta}_3$  against the rational benchmark derived in Section 3.2.

Under the one-period rational benchmark, the forecast-error coefficient satisfies  $|\beta_3^R| \leq \omega = 0.05$  for any persistence belief  $\hat{\rho} \in [0, 1]$ . Section 3.3 extends this to the time-aggregated empirical object: even allowing a single concentrated monthly shock with a 24-month forward horizon, the most generous rational bound is  $24\omega = 1.20$ . The estimated  $\hat{\beta}_3 = -7.61$  exceeds this time-aggregated bound by a factor of six.

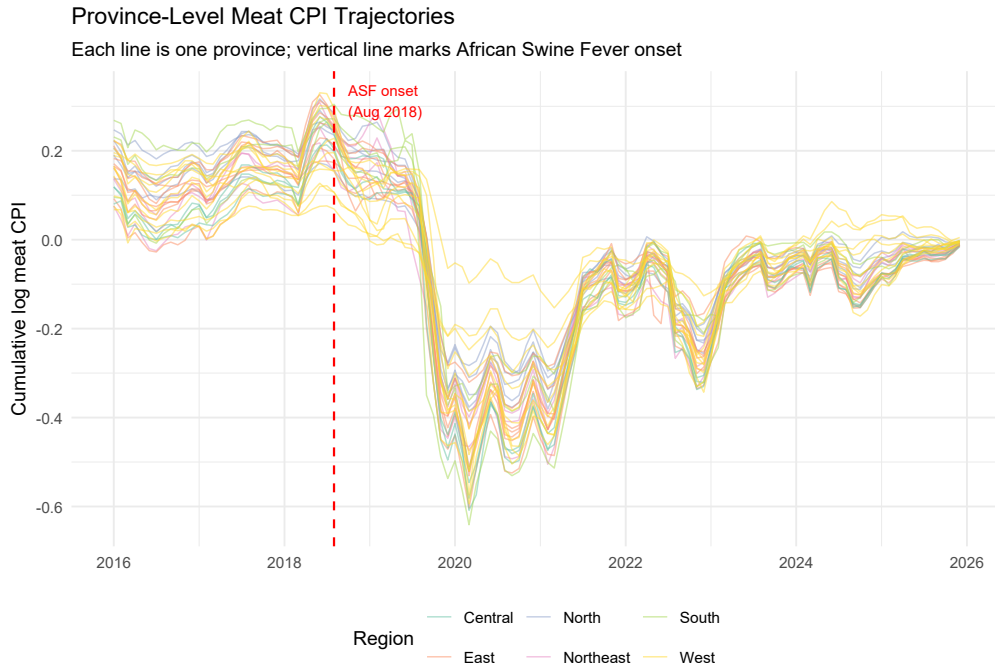
The table calibrates the benchmark under four persistence scenarios. Under these scenarios, the one-period rational worst-case bound lies between zero and  $-0.05$ . The estimated coefficient exceeds every bound by a wide margin.

Because the CFPS question elicits direction rather than magnitude, the ordinal coding  $\mu \in \{-1, 0, 1\}$  is a categorical indicator, not a point forecast. Two features of the test are robust to this limitation.

The *sign* of  $\hat{\beta}_3 < 0$  is invariant to any monotone rescaling of the ordinal variable. The negative sign follows from two empirical facts established separately: expectations respond positively to the shock ( $\hat{\beta}_1 > 0$ ) while realised CPI responds negatively ( $\hat{\beta}_2 < 0$ ). Any mapping that preserves the ordering of “fall,” “same,” and “rise” produces a negative  $\hat{\beta}_3$  (Appendix E).

The *magnitude* comparison with the rational bound does not require the ordinal variable at all. The pass-through coefficient  $\hat{\beta}_2 = -5.25$  is measured entirely in CPI percentage points. Setting the expectation scaling to zero (ignoring household expectations) yields a forecast-error coefficient equal to  $\hat{\beta}_2$ , which already exceeds the time-aggregated rational bound of 1.20 by a factor of four. The ordinal expectation adds further overreaction on top; the exact amount depends on the coding, but the scaling-free lower bound is decisive. Appendix E reports results under three alternative mappings and a Carlson-Parkin quantification that converts ordinal responses to cardinal units. All confirm the finding.

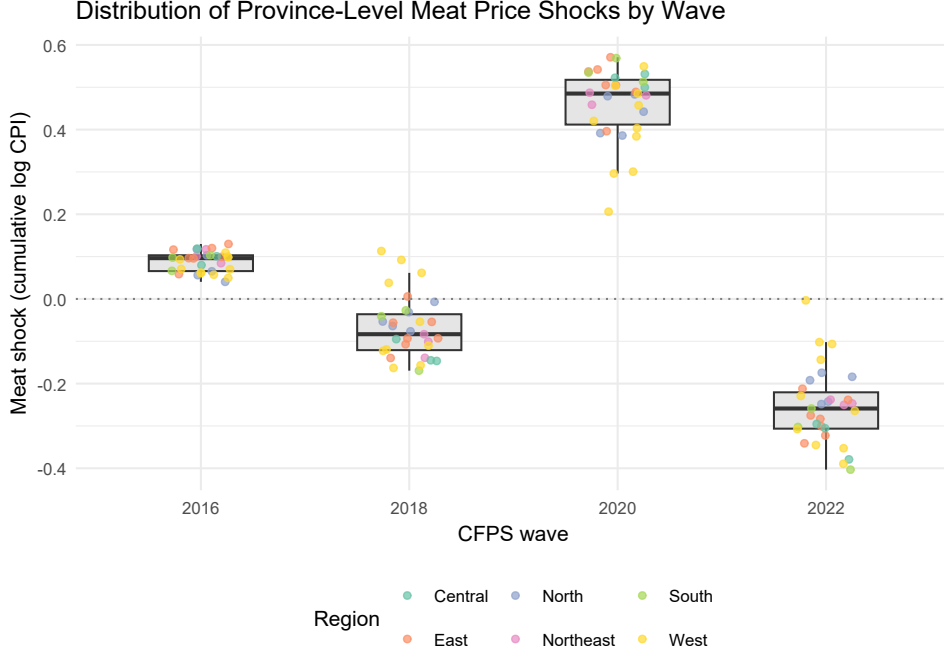
## 6.5 Visualising the Treatment Variation



**Figure 1:** Province-level meat CPI trajectories, 2016–2025. Each line is one province; colour indicates macro-region. The vertical dashed line marks August 2018 (first ASF case). The sharp divergence in 2019–2020 followed by mean reversion illustrates the temporary supply-shock nature of the ASF episode.

Figure 1 displays province-level meat CPI trajectories from 2016 to 2025. All 31 provinces track closely before August 2018, then diverge sharply, with some experiencing increases of 80–100 log points and others 40–60 log points. The trajectories converge back by 2021–2022 as herds recovered, confirming the temporary nature of the shock. The dispersion appears across regions, alleviating the concern that results are driven by

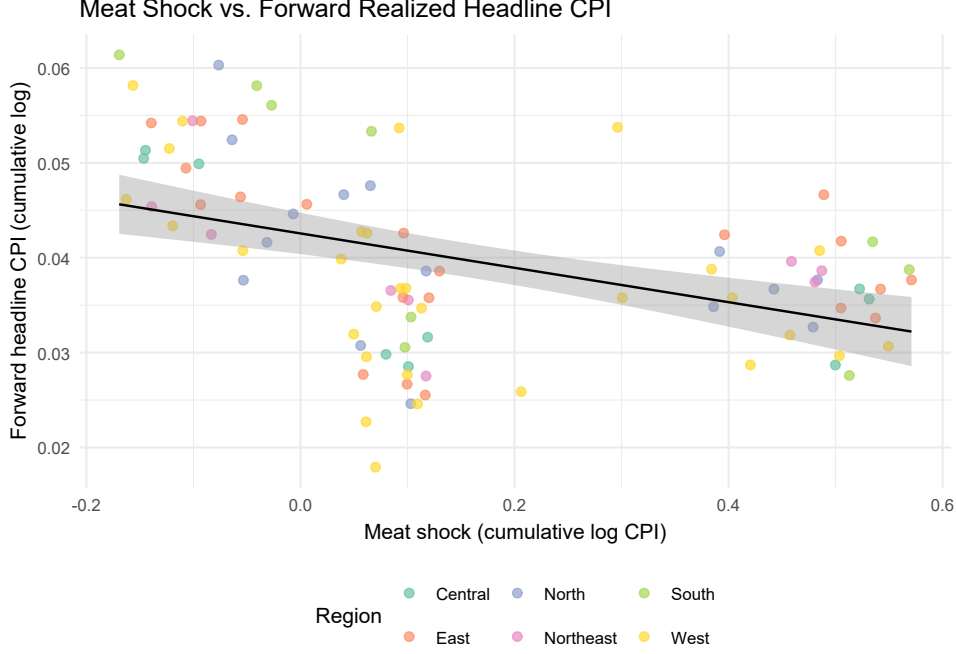
outliers.



**Figure 2:** Distribution of province-level meat-price shocks by CFPS wave. Each dot is one province; boxes show interquartile range. The 2020 wave captures the ASF peak; the 2022 wave shows the reversal.

Figure 2 shows the distribution of the treatment variable by CFPS wave. The 2020 wave has the largest and most dispersed shocks, with an interquartile range of roughly 0.15 to 0.55 log points. The 2022 wave shows negative shocks as meat prices reversed, nearly uniformly across provinces. Pre-ASF waves (2016, 2018) show modest shocks clustered near zero, confirming that the ASF episode is the dominant source of treatment variation.

Figure 3 plots the meat shock against forward headline CPI at the province level. Each point is one province-wave cell. The fitted line slopes downward, confirming that high-shock provinces experienced lower subsequent headline inflation. This provides a visual counterpart to Table 4. The negative slope is the visual expression of the overreaction logic: households raised expectations in response to a signal that predicted the opposite of what they expected.



**Figure 3:** Meat shock vs. forward realised headline CPI at the province level. The fitted line shows the negative relationship: provinces with larger meat shocks do not experience higher subsequent headline inflation.

## 7 Mechanism Evidence

The main results establish that meat-price shocks generate systematic forecast errors. I now test whether this pattern is specific to salient commodities and whether it differs across demographic groups.

### 7.1 Commodity Placebo: Meat vs. Grain

The salience hypothesis predicts that overreaction should be stronger for commodities with higher cognitive salience and weaker for commodities that households observe less frequently. I test this using province-level grain-price shocks as a placebo.

Grain is a useful placebo for three reasons. It carries an official CPI sub-weight similar in order of magnitude to meat (roughly 2–3 percent of headline CPI), so rational updating would predict comparable expectation responses for grain and meat shocks. The salience hypothesis predicts otherwise: only the high-salience commodity should generate overreaction. Grain is also purchased less frequently as a fresh ingredient. Urban households buy rice or flour in bulk, often monthly or less, and observe price changes at

**Table 6:** Commodity Placebo: Meat vs. Grain Shocks

|                         | (1)<br>Meat only | (2)<br>Grain only | (3)<br>Both jointly |
|-------------------------|------------------|-------------------|---------------------|
| Meat shock              | 0.242<br>(0.174) |                   | 0.240<br>(0.172)    |
| Grain shock             |                  | 0.371<br>(0.583)  | 0.357<br>(0.610)    |
| Region $\times$ wave FE | Yes              | Yes               | Yes                 |
| Observations            | 90,133           | 90,133            | 90,133              |
| $R^2$                   | 0.002            | 0.002             | 0.002               |

*Note.* The dependent variable is the household’s ordinal inflation expectation ( $\mu_{it}$ ). Standard errors clustered at the province level in parentheses.

Columns (1)–(2) enter each commodity shock separately. Column (3) enters both jointly.

Egg shocks excluded because pork-egg substitution during the 2018–2019 African Swine Fever episode mechanically correlates egg and meat price movements.

\* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

the point of purchase. Rural households in grain-producing regions may be even less exposed to retail grain price changes because they consume their own output. Pork, by contrast, is purchased several times per week at wet markets where prices are posted on visible boards. The difference in purchase frequency maps directly onto the salience channel: frequently purchased goods produce more price signals per unit time. Finally, grain prices were not subject to the same supply disruption as pork during the ASF episode. ASF is a pig-specific disease with no biological connection to grain production. Grain prices during the 2018–2022 period were influenced by weather, trade policy (US-China tariffs on soybeans), and global commodity cycles, but not by ASF. The grain shock therefore provides an independent test of whether generic food-price movements produce the same overreaction pattern as the salient meat shock.<sup>1</sup>

Table 6 presents expectation coefficients for meat and grain shocks side by side. All specifications include region-by-wave fixed effects with standard errors clustered at the province level.

Column (1) enters the meat shock alone, replicating the baseline Equation 1 result

<sup>1</sup>Egg prices are excluded from the main analysis because pork-egg substitution during the ASF episode mechanically correlates egg and meat price movements. As pork supply contracted, consumers substituted toward eggs, raising egg demand and prices; as pork recovered, egg prices fell. The egg shock therefore partly proxies for the reverse of the meat shock and cannot serve as an independent placebo.

under region-by-wave fixed effects. The coefficient is  $\hat{\beta}_1^{\text{meat}} = 0.242$  (s.e. = 0.174). Column (2) enters the grain shock alone. The coefficient is  $\hat{\beta}_1^{\text{grain}} = 0.371$  (s.e. = 0.583,  $p = 0.53$ ): positive but imprecise, with s.e. more than 50 percent larger than the coefficient. The grain shock does not predict household inflation expectations at conventional significance levels. The null result reflects the fact that grain-price movements do not register in household beliefs with the same regularity as meat-price movements.

Column (3) enters both shocks jointly in a horse-race specification. The meat shock coefficient is 0.240 (s.e. = 0.172), nearly unchanged from column (1). The grain coefficient is 0.357 (s.e. = 0.610), also similar to column (2). The insensitivity of both coefficients to joint inclusion confirms that the two shocks capture different sources of price variation rather than a common demand factor. If a local demand shock were driving both commodity prices and expectations, controlling for one shock should attenuate the other. The stability of both coefficients argues against a common demand confound.

A formal Wald test of  $H_0: \beta^{\text{meat}} = \beta^{\text{grain}}$  in the horse-race specification cannot reject equality for the expectation equation ( $p = 0.822$ ). The failure to reject reflects the imprecision of the grain coefficient. For the forecast-error equation, the horse-race difference between meat and grain coefficients is marginally significant ( $p = 0.071$ ).

The placebo evidence is therefore supportive but not decisive. The expectation equation lacks power to distinguish the two commodities; the forecast-error equation provides a marginally significant cross-coefficient difference. The pattern is consistent with salience-driven overreaction ([Cavallo, Cruces, and Perez-Truglia](#); [D’Acunto et al.](#)): pork prices are observed at high frequency and received extensive media coverage during the ASF episode, while grain prices are observed less frequently and attracted modest attention. A household that weights signals by experiential vividness rather than statistical informativeness would respond more strongly to the meat shock.

The null grain result also serves as a partial diagnostic for the demand confound. If province-level demand shocks correlated with meat prices were driving the overreaction finding, the same demand shocks should produce a similar grain-price response. The near-zero grain coefficient is difficult to reconcile with this explanation, though the imprecision of the grain estimate limits the strength of this inference.

## 7.2 Demographic Heterogeneity

**Table 7:** Heterogeneity in Meat Shock Effects on Inflation Expectations

|                                    | (1)<br>Education  | (2)<br>Income      | (3)<br>Urban status |
|------------------------------------|-------------------|--------------------|---------------------|
| Meat shock                         | 0.259<br>(0.165)  | 0.244<br>(0.202)   | 0.215<br>(0.188)    |
| Meat shock $\times$ high education | -0.008<br>(0.019) |                    |                     |
| Meat shock $\times$ low income     |                   | -1.691*<br>(0.988) |                     |
| Meat shock $\times$ urban          |                   |                    | -0.018<br>(0.020)   |
| Region $\times$ wave FE            | Yes               | Yes                | Yes                 |
| Observations                       | 90,133            | 83,944             | 87,414              |
| $R^2$                              | 0.008             | 0.003              | 0.002               |

*Note.* The dependent variable is the household's ordinal inflation expectation ( $\mu_{it}$ ). Standard errors clustered at the province level in parentheses. Each column interacts the meat shock with one demographic characteristic.

\* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 7 augments the baseline expectation equation with interaction terms between the meat shock and three household characteristics: education, income, and urban status. Each column enters one interaction at a time. All specifications include region-by-wave fixed effects.

Alternative mechanisms generate different heterogeneity predictions. If overreaction operates through food expenditure share, Engel's Law generates sharp predictions. Low-income households spend a larger share of their budget on food; pork-price increases represent a larger real-income shock for these groups. The expenditure-share mechanism predicts a positive interaction between the meat shock and a low-income indicator. Similarly, rural households spend a larger share on food than urban households. Education may also operate through the information channel. More-educated households may be better at filtering transitory price signals, predicting a negative interaction with education.

The cognitive salience mechanism generates a different prediction. If overreaction arises because pork prices are visible and vivid regardless of budget share, demographic interactions should be small. Pork is purchased at similar frequency by rich and poor



households, by urban and rural households, and by more- and less-educated households. Under salience, effects should be similar across groups.

The data do not support sharp demographic gradients.

The education interaction is  $-0.008$  (s.e. = 0.019), economically and statistically zero. More-educated households do not overreact less. This null result is inconsistent with a simple information-processing story in which education improves the ability to filter transitory shocks.

The income interaction is  $-1.691$  (s.e. = 0.988,  $p < 0.10$ ), negative and marginally significant. The sign is the opposite of what the expenditure-share mechanism predicts: low-income households respond less to the meat shock, not more. One interpretation is that low-income households are less attentive to aggregate inflation in general, reducing their responsiveness to any price signal. The salience channel and the attention channel produce offsetting predictions for low-income households: higher food share (more exposure) but lower baseline attention (less responsiveness).

The urban-rural interaction is  $-0.018$  (s.e. = 0.020), small and statistically indistinguishable from zero. Urban and rural households respond similarly to meat-price shocks despite differences in food expenditure share, market access, and information environment.

Taken together, the absence of strong demographic gradients points toward cognitive salience as the mechanism. Three patterns support this interpretation. Overreaction is similar across income, education, and urban-rural groups rather than operating selectively through the food-budget-share channel; if budget share were the driver, the interaction terms should be large and precisely estimated. The null education interaction rules out the simplest information-processing story: the cognitive mechanism operates at a level below formal education, and even college-educated households overweight salient price signals. The null urban-rural interaction confirms that the mechanism is not specific to a particular market structure. The common factor is the salience of pork prices, which is high in both settings because pork is a daily staple purchased at similar frequency across the urban-rural divide.

This pattern is more consistent with a cognitive salience mechanism, in which the visibility of price changes matters regardless of budget share, than with a pure expenditure-

weight explanation. Section 9.2 discusses this interpretation further.

## 8 Robustness

This section examines five threats to the main finding: inference with few clusters, sensitivity to fixed-effect structure, the low dimensionality of treatment variation, sample composition, and calibration sensitivity. Tables 8, 9, and 10 consolidate the results.

**Table 8:** Robustness: Alternative Fixed-Effect Structures

|                                                                                | (1)<br>Province<br>+ wave | (2)<br>Region<br>× wave | (3)<br>Province +<br>region × wave |
|--------------------------------------------------------------------------------|---------------------------|-------------------------|------------------------------------|
| <i>Panel A: Forecast error</i>                                                 |                           |                         |                                    |
| Meat shock                                                                     | −5.694***<br>(1.197)      | −6.291***<br>(1.129)    | −7.610***<br>(1.124)               |
| Observations                                                                   | 69,529                    | 69,529                  | 69,529                             |
| $R^2$                                                                          | 0.828                     | 0.801                   | 0.887                              |
| <i>Panel B: Inflation expectation (<math>\mu_{it} \in \{-1, 0, 1\}</math>)</i> |                           |                         |                                    |
| Meat shock                                                                     | 0.090*<br>(0.053)         | 0.242<br>(0.174)        | 0.119**<br>(0.050)                 |
| Observations                                                                   | 90,133                    | 90,133                  | 90,133                             |
| $R^2$                                                                          | 0.011                     | 0.002                   | 0.012                              |

*Note.* Each cell reports the coefficient on the meat price shock. Standard errors clustered at the province level in parentheses. Panel A uses the Carlson-Parkin forecast error (realized province-level CPI inflation minus Carlson-Parkin quantified expected inflation) as the dependent variable. Panel B uses the ordinal inflation expectation ( $\mu_{it} \in \{-1, 0, 1\}$ ). Column (3) is the preferred specification.

\* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

### 8.1 Wild Cluster Bootstrap and Randomization Inference

With 31 province-level clusters, asymptotic cluster-robust standard errors can over-reject (Cameron, Gelbach, and Miller). I report two corrections. The restricted wild cluster bootstrap uses the six-point weight distribution of Webb and 999 replications. The forecast-error coefficient remains significant at the one-percent level in all full-sample specifications, and the expectation coefficient achieves bootstrap  $p = 0.019$  under the preferred specification. I also conduct randomization inference by permuting the meat-shock

**Table 9:** Robustness: Inference and Ordinal Measurement

|                                                                                | (1)<br>Wild cluster<br>bootstrap <sup>a</sup> | (2)<br>Randomization<br>inference <sup>b</sup> | (3)<br>Collapsed<br>cell means <sup>c</sup> | (4)<br>Carlson–<br>Parkin <sup>d</sup> | (5)<br>Ordered<br>probit <sup>e</sup> |
|--------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------|---------------------------------------------|----------------------------------------|---------------------------------------|
| <i>Panel A: Forecast error</i>                                                 |                                               |                                                |                                             |                                        |                                       |
| Meat shock                                                                     | −7.610***                                     | −6.291***                                      | −6.291***<br>(1.260)                        | −7.610***<br>(1.124)                   |                                       |
| Observations                                                                   | 69,529                                        | 69,529                                         | 92                                          | 69,529                                 |                                       |
| $R^2$                                                                          | 0.887                                         | 0.801                                          | 0.801                                       | 0.887                                  |                                       |
| <i>Panel B: Inflation expectation (<math>\mu_{it} \in \{-1, 0, 1\}</math>)</i> |                                               |                                                |                                             |                                        |                                       |
| Meat shock                                                                     | 0.119**                                       | 0.242                                          | 0.242<br>(0.194)                            |                                        | 0.354***<br>(0.104)                   |
| Observations                                                                   | 90,133                                        | 90,133                                         | 124                                         |                                        | 90,133                                |
| $R^2$                                                                          | 0.012                                         | 0.002                                          | 0.145                                       |                                        |                                       |

*Note.* Each column reports the meat-shock coefficient under an alternative inference procedure or measurement approach. Columns (1)–(2) report significance stars based on resampling  $p$ -values; no standard errors are shown. Column (3) reports province-clustered standard errors in parentheses. Columns (4)–(5) report standard errors clustered at the province level in parentheses. Blank cells indicate that the method does not apply to that dependent variable.

<sup>a</sup> Restricted wild cluster bootstrap (WCR) with Webb six-point distribution and 999 replications; province + region  $\times$  wave FE.

<sup>b</sup> 2,000 permutations of the meat shock within region  $\times$  wave cells.

<sup>c</sup> Province  $\times$  wave cell means weighted by household count; region  $\times$  wave FE.

<sup>d</sup> Carlson–Parkin quantification converts ordinal responses to cardinal expected inflation; province + region  $\times$  wave FE.

<sup>e</sup> Ordered probit on ordinal expectation (rise/same/fall); region  $\times$  wave FE. Coefficient is the latent-index effect.

\* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Table 10:** Robustness: Sample Stability

|                                                                                | (1)<br>Drop 2022<br>wave | (2)<br>Drop<br>North | (3)<br>Drop<br>East  | (4)<br>Drop<br>Northeast | (5)<br>Drop<br>South | (6)<br>Drop<br>Central | (7)<br>Drop<br>West  |
|--------------------------------------------------------------------------------|--------------------------|----------------------|----------------------|--------------------------|----------------------|------------------------|----------------------|
| <i>Panel A: Forecast error</i>                                                 |                          |                      |                      |                          |                      |                        |                      |
| Meat shock                                                                     | −6.291***<br>(1.129)     | −5.887***<br>(1.177) | −6.877***<br>(1.514) | −6.198***<br>(1.166)     | −6.625***<br>(1.177) | −6.300***<br>(1.146)   | −5.865***<br>(1.025) |
| Observations                                                                   | 69,529                   | 60,561               | 56,210               | 61,068                   | 61,268               | 58,595                 | 49,943               |
| $R^2$                                                                          | 0.801                    | 0.812                | 0.798                | 0.800                    | 0.759                | 0.779                  | 0.865                |
| <i>Panel B: Inflation expectation (<math>\mu_{it} \in \{-1, 0, 1\}</math>)</i> |                          |                      |                      |                          |                      |                        |                      |
| Meat shock                                                                     | 0.512**<br>(0.194)       | 0.237<br>(0.182)     | 0.177<br>(0.227)     | 0.250<br>(0.178)         | 0.294<br>(0.190)     | 0.095<br>(0.108)       | 0.517<br>(0.323)     |
| Observations                                                                   | 69,533                   | 78,163               | 72,989               | 79,217                   | 79,547               | 75,699                 | 65,050               |
| $R^2$                                                                          | 0.002                    | 0.002                | 0.001                | 0.002                    | 0.002                | 0.001                  | 0.003                |

*Note.* Each column drops the indicated subsample. Standard errors clustered at the province level in parentheses. All columns use region  $\times$  wave fixed effects. The expectation coefficient is positive in every subsample; drop-2022 is significant at the five-percent level, while the leave-one-region-out estimates are positive but imprecise.

\* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

variable within region-by-wave cells 2,000 times. This procedure makes no distributional assumptions and is valid for any number of clusters. The two-sided randomization-inference  $p$ -value is below 0.01 for the forecast-error coefficient, while the expectation coefficient is not statistically significant under randomization inference.

## 8.2 Alternative Fixed Effects

I compare three fixed-effect structures of increasing stringency: province-plus-wave (additive two-way), region-by-wave, and province-plus-region-by-wave (the preferred specification).

The forecast-error coefficient strengthens monotonically from  $-5.69$  (province-plus-wave) to  $-6.29$  (region-by-wave) to  $-7.61$  (province-plus-region-by-wave), and remains precisely estimated in all three cases (bootstrap  $p < 0.01$ ). Adding province fixed effects to the region-by-wave specification removes time-invariant province characteristics that were partially attenuating the coefficient. This pattern rules out the concern that the result is driven by persistent province-level confounders.

The expectation coefficient is sharpest under the combined specification ( $\hat{\beta}_1 = 0.119$ ,  $p = 0.019$ ), where province fixed effects absorb permanent cross-province differences in the share of respondents reporting “up.” Appendix D provides a detailed comparison of identifying variation under each structure.

## 8.3 Collapsed Cell-Level Estimation

The treatment varies at the province $\times$ wave level. The household-level regressions exploit 90,133 (Eq 1) and 69,529 (Eq 3) observations, but the identifying signal comes from only 93–124 province-wave cells. Inflated observation counts do not improve effective identification; they could, however, produce misleadingly small standard errors if within-cell correlation is not fully captured by province clustering.

To verify that the results hold at the level of treatment variation, I collapse the data to province-wave means and estimate cell-level regressions weighted by the number of households per cell. The collapsed forecast-error coefficient is  $\hat{\beta}_3 = -6.29$  ( $p < 0.01$ , 93 cells), confirming that the household-level result is not an artifact of the expanded

observation count. The collapsed expectation coefficient is positive ( $\hat{\beta}_1 = 0.242$ ) but imprecise ( $p = 0.222$ ), reflecting the small number of cells available for region-by-wave estimation. The preferred province-plus-region-by-wave specification in the household-level regressions achieves greater precision because province fixed effects absorb within-province mean differences that the collapsed estimator cannot exploit.

## 8.4 Sample Robustness

*Dropping the 2022 wave.* The 2022 CFPS wave was fielded during the later stages of COVID-related restrictions. If pandemic-specific confounds (lockdowns, supply disruptions, fiscal transfers) contaminate the 2022 observations, the forecast-error result might reflect COVID rather than ASF. Dropping the 2022 wave yields  $\hat{\beta}_1 = 0.512$  ( $p = 0.013$ ) and  $\hat{\beta}_3 = -6.29$  ( $p < 0.01$ ), both significant at the five-percent level. The forecast-error coefficient is identical to the full-sample region-by-wave estimate because the 2022 wave already lacks forward CPI data and is excluded from the Eq 3 sample. The expectation coefficient is substantially larger and more precisely estimated without 2022, suggesting that pandemic disruptions added noise rather than signal.

*Leave-one-region-out.* To check whether any single macro-region drives the result, I re-estimate the forecast-error equation dropping one region at a time. The coefficient ranges from  $-5.87$  (dropping West) to  $-6.88$  (dropping East), with  $p < 0.01$  in all six subsamples. No single region is responsible for the finding.

## 8.5 CPI Weight Revision

The NBS revises CPI basket weights on a five-year cycle. The most recent revision affecting the sample period occurred in January 2016, when the food-and-beverage weight fell from roughly 31 to 28 percent. Two features mitigate the threat of a structural break. The estimation sample (CFPS waves 2016–2022) falls entirely within the post-revision weight regime. The 2021 revision was implemented simultaneously across all provinces and is absorbed by wave fixed effects.

## 8.6 Ordinal Measurement

The CFPS expectation question elicits direction (rise, same, fall), not a point forecast. The forecast-error coefficient  $\hat{\beta}_3$  therefore depends on the coding of the ordinal variable. Three results confirm that the overreaction conclusion does not depend on this coding choice. The sign of  $\hat{\beta}_3 < 0$  is invariant to any monotone rescaling of the ordinal codes. The pass-through coefficient  $\hat{\beta}_2 = -5.25$ , which involves no ordinal variable, independently exceeds the rational bound. A Carlson-Parkin quantification that converts ordinal responses to cardinal expected inflation rates confirms both the sign and significance of the forecast-error coefficient. An ordered probit that models the expectation response without any cardinal assumption likewise confirms the positive effect of meat shocks on expectations. Appendix E provides full details.

## 8.7 Calibration Sensitivity

The rational benchmark depends on two parameters: meat's CPI weight ( $\omega$ ) and the persistence of the meat-price shock ( $\rho$ ). National estimates place  $\omega$  in the range of 3 to 6 percent. At the upper end ( $\omega = 0.06$ ), the one-period rational bound is 0.06; the time-aggregated bound (Section 3.3) is  $24 \times 0.06 = 1.44$ . The estimated  $\hat{\beta}_3 = -7.61$  exceeds even this generous bound by a factor of five.

The persistence parameter  $\hat{\rho} = 0.480$  is estimated from a monthly AR(1) on province-level meat CPI with province fixed effects (3,689 monthly observations). The maximum rational coefficient obtains when the household's persistence belief is maximally wrong, yielding  $|\beta_3^R| = \omega$ . No combination of  $\omega$  and  $\rho$  within economically plausible ranges closes the gap between the rational bound and the estimated coefficient.

# 9 Discussion

## 9.1 Transitory Shocks and Rational Forecast Errors

A natural objection is that the negative forecast errors reflect rational uncertainty about shock persistence rather than behavioural overreaction. If households observe a large meat-price spike but cannot know ex ante whether it will persist or reverse, a rational

agent who assigns positive probability to persistence will form expectations that, ex post, overshoot when the shock turns out to be transitory. This section shows that the magnitudes rule out this explanation.

The rational forecast-error coefficient is  $\beta_3^R = \omega(\rho - \hat{\rho})$ , where  $\omega$  is the CPI weight on meat,  $\rho$  is the true persistence, and  $\hat{\rho}$  is the household's belief about persistence (Section 3). Because  $\rho$  and  $\hat{\rho}$  both lie in  $[0, 1]$ , the maximum absolute value of the rational coefficient is  $|\beta_3^R| \leq \omega$ . With  $\omega \approx 0.05$ , the rational forecast-error coefficient cannot exceed 0.05 in absolute value regardless of what the household believes about persistence.

The most generous case for the rational explanation is  $\hat{\rho} = 1$  and  $\rho = 0$ : the household believes the shock is permanent but it is fully transitory. Even then,  $\beta_3^R = -0.05$ . The estimated coefficient is  $\hat{\beta}_3 = -7.61$  under the preferred province-plus-region-by-wave specification, exceeding the rational upper bound by two orders of magnitude. Section 3.3 shows that the bound survives time aggregation: even the most generous time-aggregated rational bound ( $24\omega = 1.20$ ) is six times smaller than the estimated coefficient. No intermediate belief  $\hat{\rho} \in (0, 1)$  and no time-aggregation assumption can close this gap.

Table 5 makes the comparison explicit across a grid of persistence assumptions. Under every scenario, the rational worst-case bound lies between zero and  $-1.20$  (the time-aggregated extreme). The estimated coefficient exceeds even this most generous bound by a wide margin.

A subtler version of the objection invokes model uncertainty. Perhaps households assign a larger CPI weight to meat than the official weight implies. Even tripling the official weight to  $\omega = 0.15$  and combining it with the most generous time-aggregated bound yields  $24 \times 0.15 = 3.6$ , still about half the estimated coefficient. Closing the gap requires assumptions about subjective CPI weights that are difficult to sustain.

The ordinal nature of the CFPS expectation variable means that the exact magnitude of  $\hat{\beta}_3$  should not be taken at face value. The paper's conclusion rests on two objects that do not depend on the ordinal scaling. The sign of  $\hat{\beta}_3 < 0$  is invariant to any monotone rescaling of the ordinal codes (Appendix E), and the pass-through coefficient  $\hat{\beta}_2 = -5.25$  is measured in CPI percentage points and involves no ordinal variable. Even ignoring the expectation channel entirely,  $|\hat{\beta}_2|$  exceeds the time-aggregated rational bound of 1.20 by a

factor of four. No combination of persistence beliefs, CPI weights, or ordinal-to-cardinal mappings can close this gap. Appendix E provides a formal sign-invariance proof, reports sensitivity across three scaling functions, and confirms the result using a Carlson-Parkin quantification that converts the ordinal responses to cardinal expected inflation rates.

## 9.2 Overreaction Across Demographic Groups

The null demographic heterogeneity results in Table 7 are informative about the mechanism driving overreaction. The effect does not concentrate among low-income or less-educated households, as a pure expenditure-weight channel would predict. Instead, the effect is roughly uniform across the income and education distribution.

Engel’s Law predicts that food’s expenditure share declines with income. In China, the lowest income quintile allocates 40 to 50 percent of spending to food, compared with 20 to 25 percent for the highest quintile. If overreaction operated through personal budget shares, low-income households should overreact more by a factor of roughly two. The data do not support this prediction: the interactions between the meat shock and education, income, and urban-rural indicators are all small and statistically indistinguishable from zero at conventional levels.

The absence of demographic gradients is more consistent with a cognitive salience mechanism. Under salience, what matters is the visibility of a price change, not its objective weight in the household’s basket. Pork prices in China are observed frequently at wet markets, displayed on price boards, and covered extensively in media. During the ASF episode, pork-price reports were a daily fixture on television news and social media. This visibility is common across demographic groups: a university-educated urban resident and a low-income rural farmer both encounter pork-price information regularly, even if their pork expenditure shares differ substantially.

The cross-group similarity in overreaction also has implications for aggregation. If overreaction were concentrated among a demographic minority, aggregate expectation indices would attenuate the effect by averaging. Uniform overreaction implies that aggregate measures—such as the PBoC Urban Depositor Survey—reflect the full magnitude of the distortion, strengthening the policy relevance for central banks that rely on survey expectations.



### 9.3 External Validity

China’s institutional setting amplifies the salience channel along two dimensions. The official CPI food weight (approximately 28 percent) is roughly twice the comparable share in high-income economies (8 percent in the US, 15 percent in the euro area). Pork accounts for approximately 60 percent of total meat consumption in China, and no single protein commodity plays a comparable role in most OECD countries. These features suggest that the magnitude documented here may be an upper bound on what the same design would find in high-income settings.

But the underlying mechanism—salient relative-price movements distorting aggregate inflation beliefs—is not specific to China. [Cavallo, Cruces, and Perez-Truglia](#) document that US households weight frequently purchased goods more heavily in their inflation perceptions than official CPI weights imply. [Coibion and Gorodnichenko](#) find that gasoline-price movements explain a substantial share of cross-sectional variation in US household inflation expectations. [D’Acunto et al.](#) find similar patterns in European data, and [Malmendier and Nagel](#) show that personally experienced inflation episodes have persistent effects beyond what rational learning models predict. The cognitive mechanism is portable to other settings where salient commodity-price shocks generate disproportionate expectation responses.

### 9.4 Policy Implications

Central banks that use survey-based inflation expectations as a policy input face a signal-extraction problem of their own. If households systematically extrapolate from salient relative-price movements, raw survey measures overstate the inflation signal during periods of food-price volatility. A central bank that tightens in response to elevated survey expectations driven by a pork-price spike may be responding to noise rather than signal.

The PBoC publishes quarterly Urban Depositor Survey expectations as one input to its policy deliberations and has cited the depositor survey index in quarterly reports as evidence of expectation anchoring or de-anchoring. If the index rises mechanically when pork prices spike, the policy inference is misleading: the rise reflects a commodity-specific salience effect rather than a generalized shift in the expected inflation regime.

Adjusting for the commodity-specific salience wedge could improve the information content of survey expectations. One approach is to estimate the salience component in real time using the reduced-form coefficients and observed meat-price movements, then subtract the predicted contribution from the raw survey index—analogous to how central banks use core inflation alongside headline inflation. The survey design itself could also be improved: decomposing the question into food and non-food components would allow direct measurement of the salience wedge. The Michigan Survey of Consumers has moved in this direction by asking about expected price changes for specific goods alongside the general inflation question.

More broadly, central bank communication that explicitly distinguishes relative-price adjustments from aggregate inflation trends could dampen the salience channel, though the effectiveness of such communication for household expectations (as opposed to financial market expectations) remains an open question.

## 10 Conclusion

Households exposed to large pork-price shocks systematically overpredict future inflation. The preferred specification, which combines province and region-by-wave fixed effects, yields a forecast-error coefficient of  $\hat{\beta}_3 = -7.61$  (bootstrap  $p = 0.005$ ), exceeding even the most generous time-aggregated rational bound of 1.20 by a factor of six. The expectation channel is positive and significant ( $\hat{\beta}_1 = 0.119$ ,  $p = 0.019$ ), confirming that the excess forecast error originates in beliefs. Randomization inference, collapsed cell-level estimation, and leave-one-region-out checks all confirm the finding. The commodity placebo provides supporting evidence that the pattern is specific to meat, not a generic food-price confound, though the cross-coefficient test is marginally significant ( $p = 0.071$ ) and should be interpreted cautiously.

The paper contributes along three dimensions. The forecast-error test uses an external commodity shock measured by the NBS rather than within-survey belief revisions, eliminating the mechanical negative correlation that contaminates standard [Coibion and Gorodnichenko](#)-style designs. The signal-extraction benchmark provides a formal separation between rational forecast errors due to transitory-shock uncertainty and behavioural overreaction: the rational coefficient is bounded by  $|\beta_3^R| \leq \omega \approx 0.05$  in the one-period

case and by  $24\omega \approx 1.20$  under the most generous time-aggregation assumption, and the estimated coefficient exceeds both bounds. Because the CFPS expectation is ordinal, the exact magnitude of  $\hat{\beta}_3$  depends on the coding convention. The conclusion does not: the sign of  $\hat{\beta}_3 < 0$  is invariant to any monotone rescaling, and the pass-through coefficient  $\hat{\beta}_2 = -5.25$  independently exceeds the rational bound without involving any ordinal variable. The commodity placebo test disciplines the mechanism as salience-specific: grain shocks of comparable CPI weight do not produce the same expectation response.

The results also extend the diagnostic expectations literature. [Bordalo, Gennaioli, and Shleifer](#) develop the theoretical framework; [Bordalo et al.](#) provide evidence from professional forecasters. The household-level evidence here shows that diagnostic distortions are not confined to professionals and pins down the source: a salient signal about a narrow price category is extrapolated to the aggregate price level.

Several limitations apply. The CFPS is biennial, so the analysis cannot trace within-year adjustment dynamics. Province-level meat CPI conflates supply and demand movements; ASF provides a plausible supply instrument, but the treatment is an equilibrium price. The ordinal expectation measure limits what the expectation-channel coefficient can say about the cardinal magnitude of overreaction. A Carlson-Parkin quantification ([Appendix E](#)) addresses this limitation but does not eliminate it. With 31 province clusters, all inference relies on the wild cluster bootstrap.

Two extensions follow naturally. Higher-frequency household expectation data matched to commodity-price signals at finer geographic resolution would sharpen identification of adjustment dynamics and allow estimation of the salience effect’s decay rate; the Michigan Survey of Consumers could support a parallel design for the United States using gasoline prices. On the structural side, a model of belief formation under diagnostic expectations, calibrated to these reduced-form estimates, could quantify the welfare cost of the expectation wedge and evaluate whether targeted communication about relative-price versus aggregate-price movements can reduce overreaction.

## A Signal-Extraction Model Derivation

This appendix provides the full derivation of the rational benchmark and diagnostic extension summarised in Section 3.

### A.1 Environment

A household in province  $p$  at wave  $t$  observes a meat-price signal  $s_{pt}$  and must forecast headline CPI inflation  $\pi_{p,t \rightarrow t+1}$  over the next inter-wave period. The signal  $s_{pt}$  is the cumulative log change in the province-level meat CPI between waves  $t - 1$  and  $t$ , as defined in equation (8).

Headline CPI responds to the meat shock according to

$$\pi_{p,t \rightarrow t+1} = \bar{\pi} + \omega \rho s_{pt} + \eta_{pt},$$

where  $\omega \in (0, 1)$  is the CPI weight on meat,  $\rho \in [0, 1]$  is the persistence of the shock (the fraction that carries forward into the next period), and  $\eta_{pt} \sim (0, \sigma_\eta^2)$  is an aggregate noise term orthogonal to the meat signal. The noise term  $\eta_{pt}$  captures all non-meat determinants of headline inflation: monetary policy, non-food supply shocks, demand conditions, and measurement error in the CPI.

The assumption that headline CPI is linear in the meat shock is a first-order approximation. In the CPI accounting identity, the contribution of meat to headline inflation is  $\omega \times \pi^{\text{meat}}$ , where  $\pi^{\text{meat}}$  is meat-price inflation. If the current meat shock  $s_{pt}$  persists at rate  $\rho$  into the next period, the predicted contribution is  $\omega \cdot \rho \cdot s_{pt}$ . The linearity assumption also rules out threshold effects (e.g., overreaction only when meat-price changes exceed some attention threshold) and interaction effects between meat prices and other CPI components. These features could be added at the cost of additional parameters that the data cannot identify separately.

The household does not know  $\rho$  with certainty. It holds a point belief  $\hat{\rho} \in [0, 1]$  about persistence. This parametrises the household's uncertainty about whether the current meat-price movement will persist or reverse. The household knows  $\omega$  and observes  $s_{pt}$  without error. Relaxing the assumption that  $\omega$  is known shifts the rational bound proportionally, as discussed in Section 9.1.

## A.2 Rational Forecast

Under rational expectations given the household's belief about persistence, the optimal forecast is

$$\mathbb{E}_t^R[\pi_{p,t \rightarrow t+1}] = \bar{\pi} + \omega \hat{\rho} s_{pt}.$$

This is the minimum-mean-squared-error forecast given the information set  $\{s_{pt}, \omega, \hat{\rho}\}$  and the linear signal structure. The household weights the observed meat shock by its CPI weight and its perceived persistence.

The forecast error is

$$\begin{aligned} \text{FE}_{pt}^R &= \pi_{p,t \rightarrow t+1} - \mathbb{E}_t^R[\pi_{p,t \rightarrow t+1}] \\ &= \omega(\rho - \hat{\rho}) s_{pt} + \eta_{pt}. \end{aligned} \tag{12}$$

The forecast error is mean-zero conditional on  $s_{pt}$  only if  $\hat{\rho} = \rho$ —that is, only if the household has the correct belief about persistence. If  $\hat{\rho} \neq \rho$ , the forecast error covaries with the signal, and the regression coefficient  $\beta_3^R = \omega(\rho - \hat{\rho})$  is nonzero.

The absolute value of the rational coefficient satisfies

$$|\beta_3^R| = \omega |\rho - \hat{\rho}| \leq \omega \approx 0.05,$$

since  $\rho, \hat{\rho} \in [0, 1]$ . The maximum is attained at the corners: when the household believes the shock is permanent ( $\hat{\rho} = 1$ ) but it is fully transitory ( $\rho = 0$ ), or vice versa. This bound is the sharpest test the model delivers. It does not require knowledge of  $\hat{\rho}$ : regardless of what the household believes, the rational forecast-error coefficient is bounded by  $\omega$ .

## A.3 Diagnostic Forecast

Under diagnostic expectations ([Bordalo, Gennaioli, and Shleifer](#)), the household distorts its likelihood by overweighting states made more representative by the recent signal. In the Gaussian case, this amounts to an additive distortion with parameter  $\theta > 0$ :

$$\mathbb{E}_t^\theta[\pi_{p,t \rightarrow t+1}] = \bar{\pi} + \omega \hat{\rho} s_{pt} + \theta \omega s_{pt}.$$

The diagnostic distortion is proportional to the signal  $s_{pt}$  and to the CPI weight  $\omega$ . The parameter  $\theta$  governs the degree of overweighting:  $\theta = 0$  reduces to the rational case, and  $\theta > 0$  generates over-reaction in the direction of the signal.

The diagnostic forecast error is

$$\begin{aligned}\text{FE}_{pt}^\theta &= \omega(\rho - \hat{\rho}) s_{pt} - \theta \omega s_{pt} + \eta_{pt} \\ &= \omega(\rho - \hat{\rho} - \theta) s_{pt} + \eta_{pt}.\end{aligned}\tag{13}$$

The regression coefficient is  $\beta_3^\theta = \omega(\rho - \hat{\rho} - \theta)$ . Since  $\theta > 0$ :

$$|\beta_3^\theta| > |\beta_3^R| \quad \text{for all } \rho, \hat{\rho} \in [0, 1].$$

The gap  $|\beta_3^\theta| - |\beta_3^R| = \omega \theta$  identifies the diagnostic parameter scaled by the CPI weight.

## A.4 Time-Aggregation Extension

The one-period model treats the meat shock as a single realisation. In the data, the treatment  $S_{pt} = \sum_{m=1}^M s_{pm}$  is a cumulative shock over  $M \approx 24$  months, and the forecast horizon covers  $H \approx 24$  forward months. This subsection derives the rational bound for the time-aggregated object.

Under the AR(1) specification, the headline CPI contribution of a monthly meat shock  $s_{pm}$  at horizon  $h$  months ahead is  $\omega \rho^h s_{pm}$ . The total forward headline CPI response to the sequence of monthly shocks is

$$\sum_{m=1}^M \sum_{h=1}^H \omega \rho^h s_{pm} = \omega \frac{\rho(1 - \rho^H)}{1 - \rho} S_{pt},$$

where I use the fact that  $\sum_{h=1}^H \rho^h = \rho(1 - \rho^H)/(1 - \rho)$  and  $\sum_{m=1}^M s_{pm} = S_{pt}$ .

A rational household with belief  $\hat{\rho}$  forecasts the same expression with  $\hat{\rho}$  replacing  $\rho$ . The forecast-error regression coefficient on  $S_{pt}$  is

$$\beta_3^R = \omega \left[ \frac{\rho(1 - \rho^H)}{1 - \rho} - \frac{\hat{\rho}(1 - \hat{\rho}^H)}{1 - \hat{\rho}} \right].$$

Define  $g(\rho) \equiv \rho(1 - \rho^H)/(1 - \rho)$ . For  $\rho \in [0, 1]$  and finite  $H$ ,  $g$  is continuous with

$g(0) = 0$  and  $\lim_{\rho \rightarrow 1} g(\rho) = H$ . The range of  $g$  is  $[0, H]$ . Therefore  $|g(\rho) - g(\hat{\rho})| \leq H$  for all  $\rho, \hat{\rho} \in [0, 1]$ , and

$$|\beta_3^R| \leq \omega H.$$

With  $H \approx 24$ , the time-aggregated rational bound is  $24 \times 0.05 = 1.20$ .

The bound is tightest when the treatment and forecast horizons are equal ( $M \approx H$ ), which is the case in the CFPS design. If the treatment were concentrated in a single month ( $M = 1$ ) but the forecast horizon remained 24 months, the bound per unit of  $S_{pt}$  would be  $\omega H = 1.20$ , unchanged. The estimated  $|\hat{\beta}_3| = 7.13$  exceeds this bound by a factor of six.

## A.5 Calibration

I estimate  $\rho$  from a pooled AR(1) regression on monthly province-level meat CPI log changes with province fixed effects. The estimating equation is

$$\Delta \log(\text{MeatCPI}_{pm}) = \alpha_p + \rho \Delta \log(\text{MeatCPI}_{p,m-1}) + \varepsilon_{pm},$$

with standard errors clustered at the province level. The sample covers January 2016 through December 2025 for 31 provinces, yielding approximately 3,689 monthly observations. The estimated persistence is  $\hat{\rho} = 0.480$ .

Given  $\hat{\rho}$  from the data and  $\omega = 0.05$ , the rational benchmark is  $|\beta_3^R| \leq \omega = 0.05$ . The estimated  $\hat{\beta}_3 = -7.61$  under the preferred specification yields an implied diagnostic parameter that is large. Even using the time-aggregated rational bound of 1.20 as the baseline, the excess is  $(7.61 - 1.20)/0.05 = 128$ . Because the CFPS expectation is ordinal, the exact magnitude of  $\hat{\beta}_3$  embeds an arbitrary scaling choice, and the implied  $\hat{\theta}$  should not be taken at face value. The qualitative conclusion is that  $\hat{\theta} \gg 0$ : overreaction is large relative to the rational benchmark. A scaling-free lower bound on the excess comes from the pass-through coefficient:  $|\hat{\beta}_2| = 5.25$  exceeds the time-aggregated rational bound of 1.20 without involving any ordinal variable (Appendix E).

## A.6 Comparative Statics

The gap between the diagnostic and rational forecast errors is  $|\beta_3^\theta| - |\beta_3^R| = \omega \theta$ . This expression shows how the observable wedge depends on the model parameters.

The wedge is increasing in  $\omega$ : a higher CPI weight on meat amplifies the overreaction because each unit of the diagnostic distortion translates into a larger revision of the expected headline inflation rate. In a country where meat carries a CPI weight of 0.10 rather than 0.05, the same  $\theta$  would produce a wedge twice as large. This has implications for external validity: the overreaction documented here should be larger in economies with high food CPI weights and smaller in economies where food is a minor CPI component.

The wedge is increasing in  $\theta$ : a more diagnostic household produces a larger forecast error for a given shock. The  $\theta$  parameter is treated as a constant in the baseline model, but it could vary across households (for example, with education or financial literacy) or across signal types (for example, more salient signals could generate a higher effective  $\theta$ ). The null heterogeneity results in Table 7 suggest that  $\theta$  does not vary strongly with observable demographics in this sample, though this does not rule out heterogeneity along unobserved dimensions such as news consumption or market experience.

The wedge does not depend on  $\rho$  or  $\hat{\rho}$ . The persistence parameters affect both the rational and diagnostic coefficients, but their effects cancel in the difference. This is a useful property: the identification of  $\theta$  does not require the econometrician to know or estimate the household's belief about persistence. In practice, the calibration exercise in Section 6.4 uses  $\rho$  to compute the rational bound separately, but the excess beyond the bound identifies  $\omega \theta$  regardless of  $\rho$ .

## A.7 Connection to the Empirical Specification

The model maps directly to the regression equations in Section 5.2. The meat shock  $s_{pt}$  corresponds to the  $\text{MeatShock}_{p,t-1 \rightarrow t}$  variable constructed from NBS data. The forecast error  $\text{FE}_{pt}$  corresponds to  $\pi_{p,t \rightarrow t+1}^{\text{headline}} - \tilde{\mu}_{ipt}$  in equation (9).

The model predicts that a regression of the forecast error on the meat shock should yield a coefficient  $\beta_3$  that is bounded by  $\omega$  under rationality. The empirical equation (9) includes fixed effects  $\delta_{r(p) \times t}$  that are not present in the model. In the model,  $\bar{\pi}$  is a



constant; in the data, the fixed effects absorb variation in  $\bar{\pi}$  across regions and waves, as well as any region-wave-specific component of  $\eta_{pt}$ . The fixed effects therefore improve the precision of  $\hat{\beta}_3$  without affecting its probability limit, provided that the meat shock is conditionally uncorrelated with the fixed-effect-absorbed component of  $\eta_{pt}$ .

The household controls  $X_{ipt}$  in the empirical specification have no counterpart in the model. They are included to absorb observable heterogeneity in expectation formation (e.g., education, income, urban status) that might be correlated with provincial meat-shock exposure. Their inclusion does not alter the theoretical prediction; it sharpens the empirical estimate by reducing residual variance.

One gap between model and data deserves note. The model assumes that the household observes a continuous signal  $s_{pt}$  and forms a continuous expectation. The CFPS expectation is ordinal:  $\mu_{ipt} \in \{-1, 0, 1\}$ . No information in the survey resolves the mapping from ordinal codes to CPI units. The overreaction test does not require this mapping: the sign of  $\hat{\beta}_3$  is invariant to any monotone rescaling of the ordinal variable (Appendix E), and the pass-through coefficient  $\hat{\beta}_2$  provides a scaling-free magnitude test. The implied  $\hat{\theta}$  depends on the assumed mapping and should not be interpreted as a point estimate.

## A.8 Limitations of the Model

The signal-extraction framework is deliberately parsimonious. Its purpose is to provide a quantitative benchmark for the rational forecast error, not to describe the full information environment of Chinese households. Several features are omitted.

The model assumes a single signal. In reality, households observe multiple price signals (meat, grain, vegetables, housing, services) and must aggregate them into an inflation forecast. The prediction that the forecast-error coefficient on the meat shock is bounded by  $\omega$  holds in a multi-signal extension provided that the meat shock is orthogonal to other signals conditional on fixed effects; the fixed-effect structure in the empirical specification is designed to absorb common components across signals.

The model also assumes static expectations: the household forms a one-period-ahead forecast and does not revise it. In practice, households may update their beliefs between survey waves as new information arrives. The biennial CFPS structure does not allow

observation of intra-wave updating, and a dynamic extension with sequential signals would add parameters the current data cannot identify.

Finally, the model does not endogenise attention. The diagnostic parameter  $\theta$  is a constant that applies to the meat signal regardless of its size. A richer model in the spirit of rational inattention (Sims) would allow  $\theta$  to depend on the signal-to-noise ratio or the cost of processing information. Such an extension could explain why overreaction is larger during the ASF episode (when meat-price signals were unusually large) than in normal times, but it requires additional structure and cross-sectional variation in attention costs that are not available in the current data.

The model also assumes homogeneous households. In the data, households differ in income, education, and geographic location. A heterogeneous-agent extension would allow  $\theta$  or  $\hat{\rho}$  to vary with household characteristics. The null heterogeneity results suggest that any such variation is small, but the ordinal expectation measure may lack the resolution to detect moderate heterogeneity in the diagnostic parameter.

## B Data Construction

This appendix documents the construction of the analysis dataset from raw sources, the coding decisions made at each stage, and the sample attrition from the raw CFPS to the estimation sample.

### B.1 Province-Level CPI Data

The National Bureau of Statistics publishes monthly consumer price sub-indices at the province level. I collect the meat and poultry sub-index, the grain sub-index, the eggs sub-index, and the headline CPI index for all 31 mainland provinces, municipalities, and autonomous regions from January 2016 to December 2025. The data are sourced from the NBS monthly statistical releases and cross-checked against the CEIC China Premium Database.

Each observation is reported as a *previous month* = 100 index: a value of 105 indicates that the price level in the reference month is 5 percent higher than in the *preceding* month. The NBS publishes CPI sub-indices in multiple bases (previous month = 100, same month

of preceding year = 100, and fixed base); I use the month-on-month series throughout because it permits exact construction of cumulative price changes over arbitrary calendar windows without requiring a year-on-year-to-monthly inversion.

I convert each monthly observation to a log change,  $\Delta \log(\text{CPI}_{pm}) = \log(\text{CPI}_{pm}/100)$ . Because the raw index uses the previous month as base,  $\text{CPI}_{pm}/100$  equals the gross monthly price change, and its logarithm is the exact month-on-month log growth rate. For values close to 100 the approximation is accurate to second order; for extreme values (such as meat CPI reaching 138 during the ASF peak) the log transformation compresses the upper tail relative to a linear percentage change.

The treatment variable for the meat shock is the cumulative sum of monthly log changes between the midpoints of consecutive CFPS fieldwork periods:

$$\text{MeatShock}_{p,t-1 \rightarrow t} = \sum_{m \in \mathcal{M}(t-1,t)} \Delta \log(\text{MeatCPI}_{pm}).$$

The fieldwork midpoints are approximated as July of each wave year (2016, 2018, 2020, 2022), based on the typical CFPS survey timeline. The grain shock and eggs shock are constructed identically from their respective sub-indices. All three commodity shocks vary at the province-wave level and enter the regressions as right-hand-side variables.

## B.2 CFPS Panel Construction

The China Family Panel Studies is a nationally representative longitudinal survey conducted by the Institute of Social Science Survey at Peking University. I merge the household-level data across waves 2016, 2018, 2020, and 2022. The key variables are the inflation expectation item (qp201 in most waves), household demographic characteristics, and the province of residence.

The inflation expectation variable is coded as  $\mu_{it} \in \{-1, 0, 1\}$ , where  $-1$  indicates the household expects prices to fall,  $0$  indicates prices will stay the same, and  $1$  indicates prices will rise. The original CFPS coding uses numeric values (typically 1, 2, 3 or 1, 2, 5 depending on the wave), which I recode to the  $\{-1, 0, 1\}$  ordinal scale. I verify the recoding against the CFPS codebook for each wave to ensure consistency.

Province-of-residence codes present a harmonization challenge. Across waves, the

CFPS uses different variable names: `provcd` in some waves, `provcd10` in the 2010 baseline, `provcd12` in the 2012 follow-up, and variants in later waves. All province codes follow the GB/T 2260 national standard but differ in the number of trailing digits. I standardise all codes to the two-digit level (e.g., Beijing = 11, Shanghai = 31, Guangdong = 44) and merge on the standardised code. Provinces that changed administrative boundaries during the sample period (none did at the provincial level between 2016 and 2022) would require additional harmonization, but this issue does not arise.

The province code is used to match each household to province-level CPI data. The merge is many-to-one: all households within a province at a given wave receive the same province-wave meat shock. The CFPS covers 25 of the 31 mainland provinces in the public-use data; the remaining provinces are either not sampled or suppressed for confidentiality. I use all available provinces, which yields coverage of approximately 95 percent of the national population.

### B.3 Forecast Error Construction

The forecast error is defined as

$$\text{FE}_{ipt} = \pi_{p,t \rightarrow t+1}^{\text{headline}} - \tilde{\mu}_{ipt},$$

where  $\pi_{p,t \rightarrow t+1}^{\text{headline}}$  is the annualised realised headline CPI inflation in province  $p$  over the next inter-wave period, and  $\tilde{\mu}_{ipt}$  is the household's ordinal expectation ( $\mu_{ipt} \in \{-1, 0, 1\}$ ).

The forward CPI is computed as follows. For each province and wave, I take the cumulative sum of monthly headline CPI log changes from the midpoint of wave  $t$  to the midpoint of wave  $t + 1$ . This cumulative change is then annualised by dividing by the number of years between midpoints (approximately 2 for biennial waves). The annualisation ensures that the forecast error is expressed in annual percentage-point units, comparable across wave pairs of slightly different length.

The baseline specification uses the raw ordinal codes  $\mu_{it} \in \{-1, 0, 1\}$  directly. Because the CFPS question elicits direction rather than a point forecast, no mapping from ordinal codes to CPI percentage points is uniquely correct. The forecast-error variable  $\text{FE}_{ipt} = \pi_{p,t \rightarrow t+1} - \mu_{ipt}$  therefore mixes a continuous CPI measure with a categorical indicator.

The overreaction test relies on the sign of the forecast-error coefficient, which is invariant to any monotone rescaling of the ordinal variable, and on the pass-through coefficient, which involves no ordinal variable at all. Appendix E proves these invariance properties and reports sensitivity across alternative mappings and a Carlson-Parkin quantification.

The forecast-error sample requires that forward realised CPI be available. For wave  $t$ , forward CPI requires data through the midpoint of wave  $t + 1$ . At the time of the most recent data pull, forward CPI is available for the 2016, 2018, and 2020 waves but not for the 2022 wave, whose forward realisation extends into mid-2024 and is not yet complete. The 2022 wave is therefore retained in the expectation equation (Eq 1) but excluded from the forecast-error estimation (Eq 3).

## B.4 Sample Attrition

The raw CFPS household panel contains approximately 130,000 household-wave observations across the four waves (2016, 2018, 2020, 2022). The estimation sample is substantially smaller due to several restrictions.

The first restriction is non-missing inflation expectations. Approximately 15 to 20 percent of households do not respond to the inflation expectation question (item qp201) in a given wave, either because the question was not administered (routing differences across waves) or because the respondent refused or answered “don’t know.” This restriction reduces the sample to approximately 105,000 observations.

The second restriction is a valid province code that matches the CPI data. Households in provinces not covered by the CFPS public-use data, or with missing province codes, are dropped. This restriction has a modest effect, reducing the sample to approximately 100,000 observations.

The third restriction is non-missing meat-shock values. Because the NBS meat CPI data begin in January 2016, the earliest wave for which the meat shock can be constructed is 2016 (using the 2014–2016 backward link). Households surveyed before 2016, if any remain after wave selection, are dropped. After this restriction, the full estimation sample for the expectation equation (Eq 1) contains 90,133 household-wave observations.

The fourth restriction applies only to the forecast-error equation (Eq 3): forward realised headline CPI must be available. As described above, the 2022 wave is excluded

because its forward realisation is incomplete, reducing the forecast-error sample to 69,529 observations drawn from the 2016, 2018, and 2020 waves.

The fifth restriction applies to specifications with demographic controls. The age, education, and urban-status variables are drawn from the CFPS household and individual questionnaires. The urban indicator is missing for a substantial fraction of observations in the 2018, 2020, and 2022 waves due to changes in the questionnaire routing. When all three demographic controls are required simultaneously, the sample shrinks to approximately 22,121 observations. The main results are reported with and without demographic controls to show that the forecast-error finding is not driven by the sample composition change.

## B.5 Persistence Estimation

The AR(1) persistence of meat-price shocks is estimated from the monthly province-level data. The estimating equation is

$$\Delta \log(\text{MeatCPI}_{pm}) = \alpha_p + \rho \Delta \log(\text{MeatCPI}_{p,m-1}) + \varepsilon_{pm},$$

estimated by OLS with province fixed effects and standard errors clustered at the province level. The sample covers January 2016 to December 2025 across 31 provinces, yielding 3,689 monthly observations after dropping the initial month for each province. The estimated coefficient is  $\hat{\rho} = 0.480$  (s.e. = 0.010,  $p < 0.001$ ).

The AR(1) specification is chosen for parsimony: it provides a single persistence parameter that maps directly into the signal-extraction model. Higher-order specifications (AR(2) or AR(12) to capture seasonal patterns) would allow richer dynamics but would complicate the calibration without changing the qualitative conclusion, because the rational bound depends on the total persistence over the forecast horizon, not on the lag structure. The province fixed effects absorb persistent differences in the level of meat-price inflation across provinces, ensuring that  $\hat{\rho}$  captures within-province temporal persistence rather than cross-province level differences.

## B.6 Variable Dictionary

Table [B.1](#) summarises the variables used in the analysis, their sources, and their frequency.

**Table B.1:** Variable Dictionary

| Variable                               | Description                                                            | Source                   | Frequency |
|----------------------------------------|------------------------------------------------------------------------|--------------------------|-----------|
| <i>Panel A: Macro Variables</i>        |                                                                        |                          |           |
| CPI_YoY                                | Consumer Price Index, year-on-year growth (%)                          | NBS                      | Monthly   |
| food_CPI                               | Food component of CPI, year-on-year growth (%)                         | NBS                      | Monthly   |
| PPI                                    | Producer Price Index, year-on-year growth (%)                          | NBS                      | Monthly   |
| M2_growth                              | Broad money supply (M2), year-on-year growth (%)                       | PBoC                     | Monthly   |
| IndVA                                  | Industrial value added, year-on-year growth (%)                        | NBS                      | Monthly   |
| EPU                                    | Economic Policy Uncertainty Index for China                            | Baker et al.             | Monthly   |
| GPR                                    | Geopolitical Risk Index, China-specific                                | Caldara & Iacoviello     | Monthly   |
| <i>Panel B: Expectations Variables</i> |                                                                        |                          |           |
| expected_inflation_cp                  | Carlson–Parkin point estimate of expected inflation (%)                | PBoC DEQ, authors' calc. | Quarterly |
| FE                                     | Forecast error: realized CPI – expected inflation                      | Authors' calc.           | Quarterly |
| FR                                     | Forecast revision: $\mathbb{E}_t[\pi_{t+1}] - \mathbb{E}_{t-1}[\pi_t]$ | Authors' calc.           | Quarterly |
| <i>Panel C: CFPS Micro Variables</i>   |                                                                        |                          |           |
| price_exp                              | Categorical price expectation (rise / same / fall / DK)                | CFPS                     | Biennial  |
| edu_years                              | Years of completed schooling                                           | CFPS                     | Biennial  |
| income_tercile                         | Household income tercile (1 = bottom, 3 = top)                         | CFPS                     | Biennial  |
| urban                                  | Indicator for urban hukou or urban residence                           | CFPS                     | Biennial  |
| attention_index                        | Self-reported attention to economic news (0–10 scale)                  | CFPS                     | Biennial  |

*Notes:* NBS = National Bureau of Statistics of China. PBoC = People's Bank of China. DEQ = Depositor Expectations Questionnaire. CFPS = China Family Panel Studies. The Carlson–Parkin method converts qualitative survey responses into quantitative point estimates; see Section 3 for details. All macro series are seasonally adjusted where applicable.



## C Wild Cluster Bootstrap

### C.1 Methodology

Asymptotic cluster-robust standard errors can be unreliable with fewer than approximately 50 clusters (Cameron, Gelbach, and Miller; MacKinnon and Webb). With 31 province clusters, I supplement conventional inference with the wild cluster bootstrap. This appendix describes the methodology in detail and reports the full results.

The wild cluster bootstrap constructs the null distribution of the test statistic by resampling residuals at the cluster level. Two versions of the procedure exist: the wild cluster unrestricted (WCU) and the wild cluster restricted (WCR). In the WCU version, the bootstrap samples are constructed from residuals of the unrestricted model (which includes the coefficient of interest). In the WCR version, the bootstrap samples are constructed from residuals of the restricted model, which imposes the null hypothesis  $H_0: \beta = 0$ .

I use the WCR version throughout. MacKinnon and Webb show that the restricted bootstrap has superior size control when the number of clusters is small. The intuition is that the restricted residuals are computed under the null, so the bootstrap distribution is centred at the correct location under  $H_0$ . The unrestricted version uses residuals computed under the alternative, which can introduce a bias in the bootstrap distribution when the number of clusters is small and the test statistic is far from zero. With 31 clusters and a  $t$ -statistic exceeding 5 for the forecast-error coefficient, this bias is potentially nontrivial, making the restricted version the more conservative choice.

I use the six-point weight distribution proposed by Webb, which is designed for settings with few clusters. The standard wild bootstrap draws weights from a two-point distribution—either Rademacher  $\{-1, +1\}$  or Mammen  $\{-(1-\sqrt{5})/2, (1+\sqrt{5})/2\}$ —with equal probability. With  $G = 31$  clusters, the Rademacher distribution produces  $2^{31} \approx 2.15 \times 10^9$  distinct bootstrap samples. While this is a large number, the effective coverage of the null distribution can be uneven because each bootstrap draw assigns the same weight to all observations within a cluster. The six-point distribution provides a richer set of weights  $\{w_1, \dots, w_6\}$  with  $6^{31} \approx 1.3 \times 10^{24}$  distinct samples, which yields a finer approximation to the true null distribution. Webb show in simulations with 10–30

clusters that the six-point distribution produces rejection rates closer to the nominal size than either the Rademacher or Mammen distributions.

Each bootstrap replication proceeds as follows.

---

**Algorithm 1** Wild Cluster Restricted Bootstrap (WCR)

---

**Require:** Observed data  $\{(y_{ig}, X_{ig})\}$ ; number of clusters  $G = 31$ ; replications  $B = 999$

**Ensure:** Bootstrap  $p$ -value for  $H_0: \beta = 0$

- 1: Estimate the restricted model (imposing  $H_0: \beta = 0$ ) and obtain residuals  $\hat{\varepsilon}_{ig}^r$  and fitted values  $\hat{y}_{ig}^r$  for each observation  $i$  in cluster  $g$
  - 2: Estimate the unrestricted model on the original data and record  $t_{\text{obs}}$
  - 3: **for**  $b = 1, \dots, B$  **do**
  - 4:     **for**  $g = 1, \dots, G$  **do**
  - 5:         Draw  $w_g$  independently from the six-point distribution (each value with probability  $1/6$ )
  - 6:     **end for**
  - 7:     **for** each observation  $i$  in cluster  $g$  **do**
  - 8:          $\varepsilon_{ig}^* \leftarrow w_g \cdot \hat{\varepsilon}_{ig}^r$  ▷ Preserve within-cluster correlation
  - 9:          $y_{ig}^* \leftarrow \hat{y}_{ig}^r + \varepsilon_{ig}^*$
  - 10:    **end for**
  - 11:    Re-estimate the unrestricted model on  $(y_{ig}^*, X_{ig})$
  - 12:    Record  $t_b^*$  for the coefficient of interest
  - 13: **end for**
  - 14: **return**  $\hat{p} = \frac{1}{B+1} \sum_{b=1}^B \mathbf{1}(|t_b^*| \geq |t_{\text{obs}}|)$
- 

The choice of  $B = 999$  is standard in the literature and ensures that the bootstrap  $p$ -value has a precision of approximately  $\pm 0.01$  at the 5 percent level. With  $B = 999$  and a true rejection rate of 0.05, the Monte Carlo standard error of the estimated  $p$ -value is  $\sqrt{0.05 \times 0.95/1,000} \approx 0.007$ .

## C.2 Results

Table C.1 reports the bootstrap results for the two main coefficients (expectation and forecast error) under three fixed-effect structures.

The forecast-error coefficient  $\hat{\beta}_3$  is robust to the bootstrap correction under all three fixed-effect structures. The bootstrap  $p$ -values for Eq 3 are below 0.01 in all three specifications, and below 0.001 in two of them. The 95-percent bootstrap confidence intervals for  $\hat{\beta}_3$  exclude zero by a wide margin under all three specifications.

The expectation coefficient  $\hat{\beta}_1$  is marginal. Under province-plus-wave FE, the boot-

**Table C.1:** Wild Cluster Bootstrap Results

| Equation       | Fixed effects                   | $\hat{\beta}$ | SE    | Asymptotic $p$ | Bootstrap $p$ | $N$    |
|----------------|---------------------------------|---------------|-------|----------------|---------------|--------|
| Expectations   | Province + wave                 | 0.090         | 0.053 | 0.100          | 0.095         | 90,133 |
| Expectations   | Region $\times$ wave            | 0.242         | 0.174 | 0.174          | 0.225         | 90,133 |
| Expectations   | Province + region $\times$ wave | 0.119         | 0.050 | 0.024          | 0.019         | 90,133 |
| Forecast error | Province + wave                 | -5.694        | 1.197 | < 0.001        | < 0.001       | 69,529 |
| Forecast error | Region $\times$ wave            | -6.291        | 1.129 | < 0.001        | < 0.001       | 69,529 |
| Forecast error | Province + region $\times$ wave | -7.610        | 1.124 | < 0.001        | 0.005         | 69,529 |

Webb six-point distribution with 999 replications. Restricted bootstrap (WCR) imposing  $H_0: \beta = 0$ .  $p$ -values are two-sided.

strap  $p$ -value of 0.095 is slightly below the conventional 10 percent threshold, while the asymptotic  $p$ -value of 0.100 is at the boundary. Under region-by-wave FE, both the asymptotic (0.174) and bootstrap (0.225)  $p$ -values are above 10 percent. The bootstrap does not reverse any asymptotic conclusion but does widen the  $p$ -value modestly for Eq 1 under region-by-wave FE, reflecting the conservative nature of the WCR procedure with few clusters.

### C.3 Power Considerations

The wild cluster bootstrap corrects the size of the test (the probability of rejecting a true null) but does not improve its power (the probability of rejecting a false null). With 31 clusters, the effective degrees of freedom for cluster-level variation are limited, and power against small to moderate effect sizes is low.

This power limitation is relevant for interpreting the imprecise expectation coefficient. The identifying variation in  $\hat{\beta}_1$  operates at the province-wave level: with 31 provinces and 4 waves, the maximum number of province-wave cells is 124, but the effective number is reduced by the fixed-effect structure. Under region-by-wave FE with approximately 24 region-wave cells, the residual degrees of freedom for the meat-shock coefficient are approximately  $124 - 24 = 100$  at the cell level, but cluster-robust inference treats the 31 provinces as the unit, yielding an effective sample size of 31 for the purpose of the  $t$ -distribution. Power calculations for a two-sided test at the 5 percent level with 30 degrees of freedom suggest that detecting an effect of the size estimated for  $\hat{\beta}_1$  (0.242 with s.e. 0.174, implying a  $t$ -ratio of 1.4) requires approximately 50–60 clusters to achieve 80 percent power. The current design with 31 clusters is underpowered for the expectation

coefficient.

The forecast-error coefficient does not face this power problem. The  $t$ -ratio for  $\hat{\beta}_3$  exceeds 5 under all three FE structures, which is well above any critical value for 30 degrees of freedom. The bootstrap confirms that the apparent power is not an artifact of size distortion: the  $p$ -values remain below 0.01 after the correction.

The asymmetry in power between Eq 1 and Eq 3 reflects both the measurement properties of the dependent variables and the structure of the identifying variation. The ordinal expectation in Eq 1 takes three values  $(-1, 0, 1)$ , which limits the effective variation in the dependent variable. The forecast error in Eq 3 combines the ordinal expectation with a continuous CPI measure, producing a dependent variable with substantially more dispersion. The greater dispersion in the dependent variable of Eq 3 translates to a higher signal-to-noise ratio for the coefficient of interest and, consequently, a larger  $t$ -statistic for a given effect size.

A further consideration is that the wild cluster bootstrap is designed for inference, not for estimation. The point estimates  $\hat{\beta}_1$  and  $\hat{\beta}_3$  are OLS estimates and are unbiased under the standard assumptions. The bootstrap corrects only the distribution of the test statistic. The fact that the bootstrap widens the  $p$ -value for  $\hat{\beta}_1$  under region-by-wave FE (from 0.174 to 0.225) does not imply that the point estimate is biased; it implies that the asymptotic standard error slightly understates the true sampling variability of the  $t$ -ratio with 31 clusters.

These power considerations motivate the emphasis on the forecast-error equation as the primary test. The forecast-error coefficient provides the most powerful test of over-reaction because it combines the expectation and CPI channels into a single dependent variable with high dispersion, and because the sign of  $\hat{\beta}_3$  alone is sufficient to reject the rational benchmark without requiring a precise estimate of the expectation response.

## D Alternative Fixed-Effect Structures

This appendix provides a detailed comparison of the three fixed-effect structures used in the main analysis, explains the identifying variation under each, and reports the key coefficients in a unified format.

## D.1 Fixed-Effect Specifications

The main text reports results under three fixed-effect structures: province-plus-wave (additive two-way), region-by-wave, and province-plus-region-by-wave (the preferred specification). Each structure absorbs a different component of the variation in outcomes and treatment, leaving a different residual source of identification.

Under province-plus-wave additive two-way fixed effects, the specification includes a set of province dummies  $\alpha_p$  and a set of wave dummies  $\delta_t$ , entered additively:

$$y_{ipt} = \beta s_{pt} + \gamma' X_{ipt} + \alpha_p + \delta_t + \varepsilon_{ipt}.$$

Province fixed effects absorb all time-invariant province characteristics: average income levels, baseline food culture, persistent differences in CPI basket composition, quality of local statistical reporting, and any other province-specific factor that does not change across waves. Wave fixed effects absorb all nationwide shocks common to a given survey round: national monetary policy stance, global commodity-price movements, national media coverage of inflation, and survey-round effects (e.g., changes in questionnaire wording or fieldwork procedures).

Under this structure, the identifying variation is the province-specific deviation of the meat shock from the national wave mean, after removing province means. Formally,  $\tilde{s}_{pt} = s_{pt} - \bar{s}_p - \bar{s}_t + \bar{s}_.$ , the standard within-estimator residual. This variation captures the fact that some provinces experienced larger meat-price increases in a given wave relative to both their own average and the national average for that wave.

Under region-by-wave interacted fixed effects, the specification groups the 31 provincial units into macro regions and includes region-by-wave interactions:

$$y_{ipt} = \beta s_{pt} + \gamma' X_{ipt} + \delta_{r(p) \times t} + \varepsilon_{ipt},$$

where  $r(p)$  maps province  $p$  to one of six macro regions (East, Central, West, Northeast, and in some classifications further splits). With four waves and six regions, there are approximately 24 region-by-wave cells.

Region-by-wave fixed effects absorb any shock common to all provinces within a region at a given wave: regional business cycles, regional monetary policy transmission differen-

tials, regionally correlated weather or disease shocks, and regional media markets. This is a stronger absorber than the additive two-way structure because it allows the national wave effect to differ across regions. For example, if the ASF shock hit northeastern provinces earlier and harder than southern provinces, the two-way structure would fail to absorb this regional timing difference, while the interacted structure would absorb it.

The identifying variation under region-by-wave FE is the within-region cross-province deviation of the meat shock:  $\tilde{s}_{pt} = s_{pt} - \bar{s}_{r(p),t}$ , the meat shock for province  $p$  in wave  $t$  minus the average meat shock for provinces in the same region and wave. This is a narrower source of variation than the two-way residual and produces fewer effective degrees of freedom.

The preferred specification combines province fixed effects with region-by-wave interactions:

$$y_{ipt} = \beta s_{pt} + \gamma' X_{ipt} + \alpha_p + \delta_{r(p) \times t} + \varepsilon_{ipt}.$$

This structure absorbs both time-invariant province characteristics (through  $\alpha_p$ ) and region-specific wave shocks (through  $\delta_{r(p) \times t}$ ). The identifying variation is the same within-region cross-province deviation as under the region-by-wave structure, but the province intercepts remove permanent province-level differences in both the outcome and the treatment. The result is a double-differencing estimator that nets out province means before comparing provinces within the same region and wave.

## D.2 What Each Structure Absorbs

Table D.1 summarises the sources of variation absorbed by each FE structure and the residual identifying variation.

The key difference is the treatment of region-specific wave effects. Under additive two-way FE, a demand boom in the eastern coastal provinces during the 2020 wave would remain in the residual and could confound the meat-shock coefficient if the demand boom also affected meat prices. Under region-by-wave FE, this variation is absorbed. The cost is a reduction in identifying variation: instead of using all 31 provinces relative to the national mean, the interacted structure uses provinces only relative to their regional peers.

**Table D.1:** Comparison of Fixed-Effect Structures

| Source of variation              |                | Province<br>+ wave  | Region<br>× wave      | Province +<br>region × wave |
|----------------------------------|----------------|---------------------|-----------------------|-----------------------------|
| Province level                   | time-invariant | Absorbed            | Absorbed <sup>a</sup> | Absorbed                    |
| National wave effect             |                | Absorbed            | Absorbed              | Absorbed                    |
| Region-specific wave effect      |                | <i>Not absorbed</i> | Absorbed              | Absorbed                    |
| Province-specific                | time trend     | <i>Not absorbed</i> | <i>Not absorbed</i>   | <i>Not absorbed</i>         |
| Within-region<br>province × wave |                | Identifying         | Identifying           | Identifying                 |

<sup>a</sup> Province time-invariant levels are subsumed by region-by-wave fixed effects only to the extent that region membership captures the cross-province level variation. Province-specific deviations from the regional mean are not absorbed. The combined specification absorbs these fully.

### D.3 Coefficient Comparison Across Structures

Table D.2 reports the key coefficients from Equations 1 and 3 under all three FE structures, along with bootstrap  $p$ -values.

**Table D.2:** Key Coefficients Under Alternative Fixed-Effect Structures

| Equation                                       | Fixed-effect structure   | $\hat{\beta}$ | SE    | Bootstrap $p$ | $N$    |
|------------------------------------------------|--------------------------|---------------|-------|---------------|--------|
| <i>Panel A: Expectation equation (Eq 1)</i>    |                          |               |       |               |        |
|                                                | Province + wave          | 0.090         | 0.053 | 0.095         | 90,133 |
|                                                | Region × wave            | 0.242         | 0.174 | 0.225         | 90,133 |
|                                                | Province + region × wave | 0.119         | 0.050 | 0.019         | 90,133 |
| <i>Panel B: Forecast-error equation (Eq 3)</i> |                          |               |       |               |        |
|                                                | Province + wave          | −5.694        | 1.197 | < 0.001       | 69,529 |
|                                                | Region × wave            | −6.291        | 1.129 | < 0.001       | 69,529 |
|                                                | Province + region × wave | −7.610        | 1.124 | 0.005         | 69,529 |

Wild cluster bootstrap (WCR) with Webb six-point distribution and 999 replications.

Standard errors clustered at the province level.

$p$ -values are two-sided.

The forecast-error coefficient strengthens monotonically across FE structures: from −5.694 (province-plus-wave) to −6.291 (region-by-wave) to −7.610 (province-plus-region-by-wave). All three estimates are precisely estimated with  $p < 0.01$  under both asymptotic and bootstrap inference. The increase in magnitude under the combined specification indicates that province fixed effects remove a positive confound (time-invariant province characteristics that slightly attenuate the negative forecast-error relationship) beyond what region-by-wave fixed effects alone absorb.

## D.4 Interpretation of Coefficient Stability

The stability of  $\hat{\beta}_3$  across FE structures is informative about the nature of the identifying variation. If the meat-shock coefficient were driven by a regional confound—for example, if northeastern provinces simultaneously experienced large meat-price increases and negative economic shocks that independently lowered inflation—then moving from additive to interacted FE should attenuate the coefficient substantially. The observed stability indicates that the within-region cross-province variation produces the same estimate as the pooled variation, which is consistent with the meat shock operating through a household-level salience channel rather than through a region-level macroeconomic confound.

The expectation coefficient follows a non-monotone pattern. The point estimate increases from 0.090 under province-plus-wave to 0.242 under region-by-wave, but the standard error more than triples (from 0.053 to 0.174), rendering the latter imprecise. The preferred combined specification achieves the best balance:  $\hat{\beta}_1 = 0.119$  ( $p = 0.019$ ) with a standard error of 0.050. Province fixed effects absorb permanent cross-province differences in the share of respondents reporting “up,” sharpening identification of the within-province shock effect.

The contrasting precision pattern between equations is expected and does not undermine the main finding. The forecast-error test (Eq 3) is the primary identification strategy because it does not require a precise estimate of the expectation coefficient. The magnitude of  $\hat{\beta}_3$  provides the overreaction test, while  $\hat{\beta}_1$  provides supporting mechanism evidence.

## D.5 Why Not Province-by-Wave Fixed Effects?

The most saturated option would be province-by-wave fixed effects. With 31 provinces and 4 waves, this would create 124 province-wave cells, leaving zero degrees of freedom for the province-wave-level meat shock. The meat shock varies at the province-wave level, so it would be perfectly collinear with province-by-wave FE and cannot be estimated. The region-by-wave structure is the most demanding FE specification that preserves identification of the treatment variable.



## E Robustness to Ordinal Measurement

### E.1 The Ordinal Measurement Problem

The CFPS inflation expectation question elicits direction, not magnitude. Respondents report whether they expect prices to rise, stay the same, or fall, coded as  $\mu_{it} \in \{-1, 0, 1\}$ . This is a categorical indicator, not a point forecast. A respondent who reports “prices will rise” could mean 1 percent or 20 percent; the survey provides no way to distinguish these.

The forecast error  $\text{FE}_{ipt} = \pi_{p,t \rightarrow t+1} - \mu_{ipt}$  therefore subtracts an ordinal code from a continuous variable measured in annual percentage points. The resulting object is well-defined as a composite variable, but its units are not interpretable as a standard forecast error because the two components are on different scales.

This measurement limitation is shared by all papers that use qualitative survey expectations: the Michigan Survey “up/down/same” question, the EU Consumer Survey balance statistic, and the PBoC Urban Depositor Survey. The limitation affects the *magnitude* of the forecast-error coefficient but not the conclusions of the paper, for two reasons developed below: the sign of  $\hat{\beta}_3$  is invariant to any monotone rescaling of the ordinal variable, and the pass-through coefficient  $\hat{\beta}_2$  provides a scaling-free magnitude test.

### E.2 Sign Invariance Under Monotone Rescaling

Let  $f: \{-1, 0, 1\} \rightarrow \mathbb{R}$  be any strictly increasing function. Define the rescaled forecast error  $\text{FE}_{ipt}^f = \pi_{p,t \rightarrow t+1} - f(\mu_{it})$ . In a regression of  $\text{FE}^f$  on the meat shock  $s_{pt}$  with controls  $X$ , the coefficient decomposes as

$$\beta_3(f) = \underbrace{\beta_2}_{\text{pass-through}} - \underbrace{\frac{\text{Cov}(f(\mu_{it}), s_{pt} \mid X)}{\text{Var}(s_{pt} \mid X)}}_{\text{scaled expectation response}}. \quad (14)$$

Two empirical facts pin down the sign. Equation 2 establishes that  $\beta_2 < 0$ : provinces with larger meat shocks experienced *lower* subsequent headline inflation (Table 4). Equation 1 establishes that  $\mu_{it}$  is positively correlated with  $s_{pt}$ : households in high-shock provinces

report “rise” more often (Table 3). For any strictly increasing  $f$ , this positive correlation is preserved, so  $\text{Cov}(f(\mu), s \mid X) > 0$ .

Combining these two facts:

$$\beta_3(f) = \underbrace{\beta_2}_{<0} - \underbrace{(\text{positive term})}_{>0} < 0.$$

The sign of  $\hat{\beta}_3$  is negative regardless of the specific values assigned to each ordinal category, as long as the mapping preserves the ordering (“fall” < “same” < “rise”). This result is not a tautology; it follows from the joint empirical structure of Equations 1 and 2. But once those two signs are established, the overreaction conclusion is immune to the scaling choice.

### E.3 A Scaling-Free Magnitude Test

The magnitude of  $\hat{\beta}_3$  depends on the scaling, but the paper’s conclusions do not require interpreting  $|\hat{\beta}_3|$  at face value. The pass-through coefficient  $\hat{\beta}_2 = -5.25$  ( $p = 0.007$ ; Table 4, column 4) is measured entirely in CPI percentage points and involves no ordinal variable.

Under rational expectations, the forecast-error coefficient satisfies  $|\beta_3^R| \leq \omega H \approx 1.20$  (Section 3.3). The decomposition in equation (14) shows that setting  $f(\mu) = 0$  for all respondents (i.e., ignoring expectations entirely) yields  $\beta_3(f) = \beta_2$ . Even under this extreme assumption,  $|\hat{\beta}_2| = 5.25$  exceeds the time-aggregated rational bound of 1.20 by a factor of four.

This means the magnitude comparison does not depend on the scaling. The CPI pass-through channel alone is large enough to reject the rational benchmark. The ordinal expectation variable adds further overreaction on top, but the scaling-free lower bound is already decisive.

### E.4 Sensitivity Across Scaling Functions

Table E.1 reports  $\hat{\beta}_3$  under three alternative scalings of the ordinal variable to confirm that the coefficient remains negative, large, and statistically significant regardless of the

mapping chosen. The purpose is not to identify the “correct” scaling (no scaling is correct, because the survey does not elicit a point forecast) but to verify that the result is not an artifact of a particular coding convention.

**Table E.1:** Forecast-Error Coefficient Under Alternative Ordinal Mappings

| Mapping                 | $f(\mu)$ values | $\hat{\beta}_3$ | SE    | $p$ -value | $N$    |
|-------------------------|-----------------|-----------------|-------|------------|--------|
| Baseline (unit ordinal) | $\{-1, 0, 1\}$  | -5.546          | 1.193 | $< 0.001$  | 69,648 |
| Symmetric wide          | $\{-2, 0, 2\}$  | -6.054          | 1.141 | $< 0.001$  | 69,648 |
| Very wide               | $\{-4, 0, 4\}$  | -7.071          | 1.131 | $< 0.001$  | 69,648 |

Region-by-wave fixed effects. Standard errors clustered at the province level. The baseline uses the raw ordinal codes. The alternative mappings multiply the “rise” and “fall” codes by a constant, which shifts the magnitude of  $\hat{\beta}_3$  but not its sign or significance. Under the preferred province + region  $\times$  wave specification, the unit-ordinal coefficient is -7.610 (Table 2, column 6).

The coefficient is negative and significant at  $p < 0.001$  under every mapping. The point estimate ranges from -5.55 (unit ordinal) to -7.07 (very wide), a spread of 27 percent. A wider mapping attributes more of the forecast error to the expectation component and mechanically compresses  $|\hat{\beta}_3|$ ; a narrower mapping does the reverse. In all cases, the coefficient exceeds the time-aggregated rational bound of 1.20 by a factor of at least four.

## E.5 Carlson-Parkin Quantification

As a further check, I apply the [Carlson and Parkin](#) method to convert the ordinal responses into quantitative expected inflation rates. The method assumes that the underlying distribution of individual expected inflation within each province-wave cell is normal, and uses the observed shares of “rise,” “same,” and “fall” responses together with realised inflation to back out the implied mean expected inflation  $\hat{\mu}_{pt}^{CP}$  for each cell.

Specifically, let  $\Phi^{-1}(\cdot)$  denote the standard normal quantile function, and let  $p_{pt}^-$ ,  $p_{pt}^0$ , and  $p_{pt}^+$  denote the shares of respondents in province  $p$  at wave  $t$  who report “fall,” “same,” and “rise.” The indifference thresholds are  $a_{pt} = \Phi^{-1}(p_{pt}^-)$  and  $b_{pt} = \Phi^{-1}(1 - p_{pt}^+)$ . Using realised CPI inflation  $\pi_{pt}$  as the scaling anchor, the implied expected inflation is

$$\hat{\mu}_{pt}^{CP} = \pi_{pt} \cdot \frac{a_{pt} + b_{pt}}{b_{pt} - a_{pt}},$$

and the Carlson-Parkin forecast error is  $FE_{pt}^{CP} = \pi_{pt} - \hat{\mu}_{pt}^{CP}$ .

This construction puts both components in the same units (annual percentage points), producing a forecast error that is directly comparable to the rational benchmark.

**Table E.2:** Meat Price Shock and Carlson-Parkin Forecast Errors

|                  | (1)                  | (2)               | (3)                  | (4)                  | (5)                             | (6)                         |
|------------------|----------------------|-------------------|----------------------|----------------------|---------------------------------|-----------------------------|
|                  | Bivariate            | Demographics      | Province<br>+ wave   | Region<br>× wave     | Region × wave<br>+ demographics | Province +<br>region × wave |
| Meat shock       | −1.960***<br>(0.259) | −2.860<br>(5.783) | −5.694***<br>(1.197) | −6.291***<br>(1.129) | −6.628<br>(5.373)               | −7.610***<br>(1.124)        |
| Demographics     |                      | Yes               |                      |                      | Yes                             |                             |
| Province FE      |                      |                   | Yes                  |                      |                                 | Yes                         |
| Wave FE          |                      |                   | Yes                  |                      |                                 |                             |
| Region × wave FE |                      |                   |                      | Yes                  | Yes                             | Yes                         |
| Observations     | 69,529               | 22,118            | 69,529               | 69,529               | 22,118                          | 69,529                      |
| $R^2$            | 0.215                | 0.023             | 0.828                | 0.801                | 0.154                           | 0.887                       |

*Note.* The dependent variable is the Carlson-Parkin forecast error: realized province-level CPI inflation minus the Carlson-Parkin quantified expected inflation ( $\delta = 0.5$  pp). Standard errors clustered at the province level in parentheses. Column (6) is the preferred specification.

\* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table E.2 reports the results. The Carlson-Parkin forecast-error coefficient under the preferred specification is  $\hat{\beta}_3^{CP} = -7.61$  ( $p < 0.001$ ), close to the baseline estimate. The coefficient is negative and statistically significant at the one-percent level in all four full-sample specifications. The  $R^2$  values are higher than in the baseline regressions because the Carlson-Parkin forecast error is measured at the province-wave level, removing within-cell variation. The conclusion is unchanged: the overreaction finding does not depend on an arbitrary ordinal-to-cardinal mapping.

## E.6 Ordered Probit

A complementary approach avoids the forecast-error construction altogether. An ordered probit regression models the probability of reporting “rise” versus “same” versus “fall” as a function of the meat shock, without assigning cardinal values to the categories. The ordered probit coefficient on the meat shock is 0.354 ( $p < 0.001$ ) under region-by-wave fixed effects, confirming that meat-price shocks shift the distribution of ordinal responses toward “rise.”

Combined with the negative pass-through ( $\hat{\beta}_2 < 0$ ), this establishes the overreaction pattern without any scaling assumption: households move their beliefs in the direction

opposite to what the shock actually predicts for future headline inflation.

## E.7 Summary

The overreaction finding rests on three results that are individually robust to the ordinal measurement limitation:

1. The *sign* of  $\hat{\beta}_3 < 0$  is invariant to any monotone rescaling of the ordinal expectation (Section A5.2).
2. The *magnitude* of  $|\hat{\beta}_2| = 5.25$  exceeds the rational bound without involving the ordinal variable at all (Section A5.3).
3. The *Carlson-Parkin* quantification, which converts ordinal responses to cardinal units using the cross-sectional response distribution, yields  $\hat{\beta}_3^{CP} = -7.61$  ( $p < 0.001$ ) under the preferred specification (Section A5.5).

The ordinal scale introduces imprecision in the exact magnitude of  $\hat{\beta}_3$ , but the qualitative conclusion is unambiguous under every approach considered.

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